A Poisson Transition and Bias Law for KSG Mutual-Information Estimation

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October 10, 2026

Abstract

We quantify the error of the strict-radius Kraskov–Stögbauer–Grassberger (KSG-1) mutual-information estimator on a Gaussian channel as the sample size grows and the channel becomes nearly deterministic. For fixed dimension p and neighbor index k, the resolution parameter is $u_n = ne^{-I_n}$, where I_n is the true mutual information in nats. When $u_n \to u \in (0, \infty)$, we prove that the estimation error of strict-radius KSG-1 converges in mean square to an explicit limit $B_{p,k}(u)$, the estimator's limiting bias. Thus sampling fluctuations vanish, while the mean-squared error approaches $B_{p,k}(u)^2$. The bias follows from an explicit marked-Poisson law for the joint and marginal neighborhood counts. The same law gives the limiting fraction of neighborhoods that make the largest possible contribution to the estimate. This saturation probability decreases continuously and strictly from one to zero, allowing Gaussian-model bias correction on fixed transition intervals. We also characterize the neighboring count regimes and both endpoints of the bias curve. Independent numerical calculations test the limiting law against finite-sample KSG. They show substantial bias even when the true information is below the estimator's ceiling and saturation is rare. A saturation-based correction experiment reduces estimation error but shows that estimating the resolution parameter from finite-sample saturation can remain difficult.

1 Introduction

We study mutual information (MI) estimation for a Gaussian channel whose output becomes nearly determined by its input. As the channel noise vanishes, the mutual information grows and the joint distribution concentrates near the diagonal. A nearest-neighbor estimator must resolve the shrinking width of this distribution from finitely many observations. We ask how the estimation error behaves when the sample size grows at the same time as the noise vanishes.

The Kraskov–Stögbauer–Grassberger estimator (KSG) [7] measures dependence by comparing joint and marginal nearest-neighbor counts at a shared radius. For a fixed distribution, existing analyses establish consistency and bounds on bias and variance [4, 13]. Here the channel noise shrinks with sample size, changing the density, and the regularity constants in those analyses need not remain bounded. Fundamental limits on KSG estimation are well studied: the estimator has a ceiling of order $\log n$, and estimating large MI accurately can require a sample size exponential in that information [3]. The ceiling bounds what KSG can report, but does not quantify its error when the true information lies below it. Our question is therefore not whether an exponential sample requirement exists, but how KSG behaves at that scale.

Geometric methods address the local nonuniformity underlying this behavior [9]. Practical bias diagnostics and error bars [5] assess estimation reliability, but do not give the limiting error for the changing distribution studied here. Likewise, logarithmic limits on distribution-free, high-confidence information lower bounds [10] do not characterize KSG's bias. Those limits concern guarantees across distributions, rather than the error of this estimator within a specified Gaussian family. We derive such a bias law under an explicit Gaussian model. The channel's mutual information is known exactly, $I_n = (p/2)\log(1/\delta_n)$, where p is the dimension and δ_n controls the noise level. This lets us measure KSG's error against a known reference as the channel becomes nearly deterministic. The resolution parameter we use throughout is

$$u_n = ne^{-I_n} = n\delta_n^{p/2}.$$

Here I_n is measured in nats. This compares the sample size n with e^{I_n}, as the known exponential sample requirement suggests. It also has a geometric interpretation. The noise width is of order $\sqrt{\delta_n}$, while the source-neighbor spacing is of order $n^{-1/p}$, up to a local density factor. Their ratio is of order $u_n^{1/p}$. When u_n tends to a positive constant, neither scale becomes negligible relative to the other; this is the critical regime. For fixed dimension p and neighbor index k, when $u_n \to u \in (0, \infty)$, the neighborhood counts converge to an explicit mixed-Poisson law. We use this law to prove that

$$\widehat{I}_{n,k} - I_n \longrightarrow B_{p,k}(u) \text{ in } L^2,$$

where $B_{p,k}(u)$ is the computable limiting bias. Along this shrinking-noise sequence, the estimation error concentrates around $B_{p,k}(u)$, which need not be zero.

KSG chooses its radius from neighbors that are close in both X and Y. At the same radius, the X-count requires closeness only in X, and the Y-count only in Y. Both can therefore include additional observations outside the joint neighborhood. In the critical regime, rescaling distances by the noise width gives a marked Poisson process: each source point carries a Gaussian noise value that determines its output location. For fixed source and noise values of the query, this process describes the selected joint radius and the extra marginal counts. Averaging over the query values then gives the limiting count law for a randomly chosen observation.

A neighborhood is *saturated* when it makes the largest possible contribution to the estimate. The count law gives the limiting saturation fraction $q_{p,k}(u)$. This curve is continuous and strictly decreasing, approaching one as $u \to 0$ and zero as $u \to \infty$. Within the Gaussian model, this gives a bias correction: on a fixed compact critical interval, we invert the curve to estimate u from the observed saturation fraction and subtract the corresponding bias.

We also characterize the counts on either side of the critical regime. If $u_n \to 0$, a typical neighborhood saturates and the estimate grows like $\log n$. If $\delta_n \to 0$ and $u_n \to \infty$, the marginal counts grow on the $\sqrt{u_n}$ scale. Separately, in the limiting Poisson law, the small-u bias approaches the gap between the ceiling and the true information, and the bias tends to zero as $u \to \infty$.

Finite-sample experiments compare KSG's mean error and saturation probability with independently computed Poisson predictions. They show substantial error even below the estimator's ceiling, when saturation is rare. The correction reduces this error, although estimating u from finite-sample saturation can limit its accuracy.

Our results build on local Poisson approximation and geometric bounds for nearest-neighbor statistics [1, 8, 11]. We control marginal counts over regions that are unbounded in the other coordinate, and establish convergence of the full estimator despite dependence among neighborhoods. Section 2 fixes the channel and estimator. Sections 3 and 4 derive the critical count law and estimator limit. Section 5 treats the thin and regular regimes and the bias endpoints, and Section

6 derives saturation-based bias correction. Numerical results appear in Section 7. Proofs of the supporting tail, geometric, and integrability estimates appear in Appendices A–E.

2 Channel, estimator, and resolution scale

2.1 The Gaussian channel

Fix integers $p, k \geq 1$. For every $n > k$, take mutually independent standard Gaussian vectors $X_i, Z_i \in \mathbb{R}^p$, and set

$$Y_{i,n} = (1 - \delta_n)X_i + \sqrt{\delta_n(1 - \delta_n)}Z_i, \ 0 < \delta_n < 1. \tag{1}$$

The observations $(X_i, Y_{i,n})$ are iid within each sample size. The channel may change across sample sizes; no coupling of samples across n is required.

The mutual information is

$$I_n := I(X; Y_n) = \frac{p}{2} \log \frac{1}{\delta_n}. \tag{2}$$

Indeed, $\mathrm{Cov}(Y_n) = (1 - \delta_n)I_p$, while $\mathrm{Cov}(Y_n \mid X) = \delta_n(1 - \delta_n)I_p$. Taking the difference of the corresponding Gaussian differential entropies gives

$$I_n = \frac{1}{2} \log \frac{\det\{(1 - \delta_n)I_p\}}{\det\{\delta_n(1 - \delta_n)I_p\}} = \frac{p}{2} \log \frac{1}{\delta_n}.$$

As $\delta_n \to 0$, the output approaches the input and I_n diverges. All logarithms in the theory are natural. Numerical errors are divided by $\log 2$ when reported in bits.

Figure 1 illustrates the shrinking noise width around the diagonal, not an estimator neighborhood. The coordinate difference $Y - X$ has variance δ, so the standard deviation perpendicular to $Y = X$ is $\sqrt{\delta/2}$.

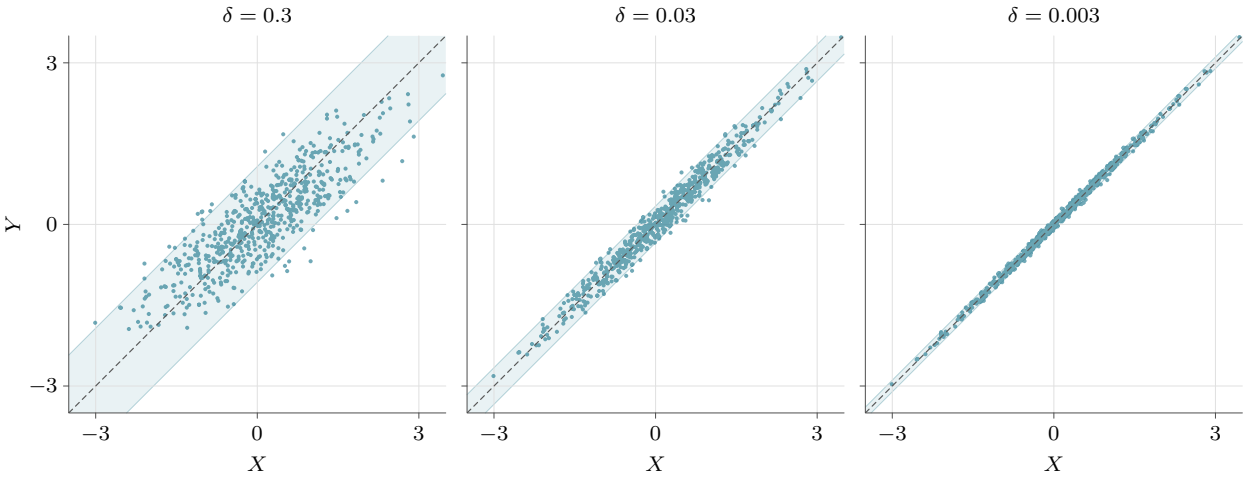


Figure 1: The scalar Gaussian channel approaches $Y = X$ as δ decreases. Each panel shows 600 pairs generated by equation (1) with $p = 1$, using the same source and noise draws across panels. Identical axis limits and equal scales on the two axes make the narrowing visible; the dashed line is $Y = X$. Shading marks $|Y - X| \leq 1.96\sqrt{\delta}$, a 95% probability band, not a hard support boundary.

2.2 Strict-radius KSG-1

For query i and competitor $j \neq i$, let

$$A_{ij} = \|X_j - X_i\|_\infty,\ B_{ij} = \|Y_{j,n} - Y_{i,n}\|_\infty,\ J_{ij} = A_{ij} \vee B_{ij}.$$

Here $a \vee b = \max(a, b)$. The joint ball of radius r is a cube in $\mathbb{R}^{2p}$: an axis-aligned square around the query when $p = 1$, not the diagonal band in figure 1. Let $\varepsilon_{i,n}$ be the kth smallest joint distance. The two marginal counts exclude the query and use a strict inequality:

$$N_{x,i} = \#\{j \neq i : A_{ij} < \varepsilon_{i,n}\},\ N_{y,i} = \#\{j \neq i : B_{ij} < \varepsilon_{i,n}\}. \tag{3}$$

The estimator is

$$\widehat{I}_{n,k} = \psi(k) + \psi(n) - \frac{1}{n}\sum_{i=1}^{n}\{\psi(N_{x,i}+1) + \psi(N_{y,i}+1)\}, \tag{4}$$

where ψ is the digamma function and $\psi(m+1) - \psi(m) = 1/m$. We use this strict-radius KSG-1 convention [7] throughout, including its norm, query exclusion, and +1 shifts.

For the Gaussian sample, each joint or marginal distance list is tie-free almost surely, and each competitor's two marginal distances are unequal. The unique kth joint neighbor is on the boundary in exactly one marginal. It is therefore excluded from one marginal count and included in the other. The first $k-1$ joint neighbors contribute to both. After possibly exchanging the marginals, every neighborhood has counts at least $(k-1, k)$, which gives

$$\widehat{I}_{n,k} \leq C_{n,k} := \psi(n) - \psi(k+1). \tag{5}$$

We call a query neighborhood *saturated* when its unordered count pair is $\{k-1, k\}$. Define

$$S_{i,n} = \mathbf{1}\{\{N_{x,i}, N_{y,i}\} = \{k-1, k\}\},\ \overline{S}_{n,k} = \frac{1}{n}\sum_{i=1}^{n} S_{i,n}.$$

The estimator equals $C_{n,k}$ exactly when every neighborhood is saturated. The order-$\log n$ ceiling and exponential sample obstruction are prior results [3]. Their two-variable algebraic ceiling agrees with equation (5), although their derivation uses a rectangle-based KSG-2 convention. Our count-pair argument and every result below concern equations (3) and (4).

2.3 Noise width and source-neighbor spacing

The noise width of the joint Gaussian law is of order $\sqrt{\delta_n}$, whereas the local source-neighbor spacing is of order $n^{-1/p}$. Their ratio leads to

$$u_n = n\delta_n^{p/2} = ne^{-I_n},\ u_n^{1/p} = \frac{\sqrt{\delta_n}}{n^{-1/p}}. \tag{6}$$

In the thin regime, $u_n \to 0$, the noise width becomes small compared with the source-neighbor spacing. In the critical regime, $u_n \to u \in (0, \infty)$, the two scales remain comparable. In the regular regime, $\delta_n \to 0$ and $u_n \to \infty$, the source-neighbor spacing becomes small compared with the noise width. These are overall scale comparisons; the local constants still depend on the query values.

3 Limiting Neighborhood Counts

A competitor can enter a source or output marginal neighborhood without entering the joint neighborhood. We construct the limiting counts in these two marginal-only regions when $u_n \to u \in (0, \infty)$.

3.1 Rescaling the Neighborhood

Condition on the query's source and noise values, $X_1 = a$ and $Z_1 = b$. For every competitor, use noise-scale coordinates

$$\xi_{j,n} = \frac{X_j - a}{\sqrt{\delta_n}},\ \eta_{j,n} = \frac{Y_{j,n} - Y_{1,n}}{\sqrt{\delta_n}}. \tag{7}$$

Subtracting the query channel equation gives the exact relation

$$\eta_{j,n} = (1 - \delta_n)\xi_{j,n} + \sqrt{1 - \delta_n}(Z_j - b). \tag{8}$$

The common rescaling preserves the joint max-norm geometry and keeps the noise displacement visible. The conditional competitor process converges to a Poisson point process on $\mathbb{R}^p \times \mathbb{R}^p$ with intensity

$$\lambda_{a,b}(s,t) = u\varphi_p(a)\varphi_p(b + t - s). \tag{9}$$

Here φ_p is the standard p-dimensional Gaussian density. After rescaling, the source points have constant intensity $u\varphi_p(a)$. A source displacement s produces an output displacement $t = s + W - b$, where $W \sim N(0, I_p)$ is that point's noise value. This explains the second factor $\varphi_p(b + t - s)$ in the intensity.

We now change units so that a cube of radius r contains $(2r)^p$ points on average in either marginal. This makes the marginal counts easier to characterize. Set $h = \{u\varphi_p(a)\}^{1/p}$ and scale both coordinates by h: $S' = hs$, $T' = ht$. The source points S' form a homogeneous Poisson process of unit intensity. Attach to each point a mark $W \sim N(0, I_p)$, independently across points and independently of the source process, and put

$$T' = S' + h(W - b). \tag{10}$$

This also gives unit intensity in the output marginal: the normalized joint intensity is $h^{-p}\varphi_p\{b + (t' - s')/h\}$, whose integral in s' is one. Thus a radius-r cube has mass $(2r)^p$ in either marginal.

A point lies in the joint cube only when both $\|S'\|_\infty \le r$ and $\|T'\|_\infty \le r$. For a fixed noise value W, each source coordinate must lie in two intervals of length $2r$, whose centers are $h|W_q - b_q|$ apart. Their intersection has length $(2r - h|W_q - b_q|)_+$, where $x_+ = \max(x, 0)$. Multiplying these lengths gives the volume of source locations satisfying both constraints. Averaging over the independent Gaussian coordinates gives the mean number of points in the joint cube

$$\Lambda_{h,b}(r) = \prod_{q=1}^{p} \mathbb{E}_W[(2r - h|W_q - b_q|)_+]. \tag{11}$$

This function is continuous and strictly increasing from zero to infinity. Indeed, the joint intensity is positive on every nonempty joint annulus, and $\Lambda_{h,b}(r)/(2r)^p \to 1$ by dominated convergence. When indexed by enclosed intensity, the joint counts form a unit-rate Poisson process. Its kth arrival has a Gamma$(k, 1)$ distribution, so the selected radius R satisfies

$$\Lambda_{h,b}(R) = G_k, \tag{12}$$

where $G_k \sim$ Gamma$(k, 1)$, with shape k and rate one. Subtracting the joint mass from either marginal mass gives

$$\mu = \mu_u = (2R)^p - G_k. \tag{13}$$

This mass is strictly positive and finite almost surely for every $u > 0$. It depends on the query's source and noise values and on G_k.

Proposition 3.1 (Critical count law)**.** *Draw mutually independent $A, B \sim N(0, I_p)$, representing the query's source and noise values, and $G_k \sim \mathrm{Gamma}(k, 1)$. Set $h = \{u\varphi_p(A)\}^{1/p}$, solve equation (12) with $b = B$, and define μ_u by equation (13). Conditional on (A, B, G_k), draw independent $P_x, P_y \sim \mathrm{Pois}(\mu_u)$ and an independent fair Bernoulli variable J. Put*

$$(M_x, M_y) = \begin{cases} (k - 1 + P_x,\ k + P_y), & J = 0, \\ (k + P_x,\ k - 1 + P_y), & J = 1. \end{cases} \tag{14}$$

If $u_n \to u \in (0, \infty)$, then

$$(X_1, Z_1, N_{x,1}, N_{y,1}) \Rightarrow (A, B, M_x, M_y).$$

In particular,

$$\mathbb{P}\{\{M_x, M_y\} = \{k - 1, k\} \mid A, B, G_k\} = e^{-2\mu_u}. \tag{15}$$

Proof. Fix the query values $(X_1, Z_1) = (a, b)$. The exact density in lemma A.2, multiplied by $n - 1$, converges locally uniformly to equation (9). Conditional independence therefore gives the Poisson process limit in lemma A.3.

Convergence on bounded joint regions is not enough: a marginal neighborhood can include a point far away in the other coordinate. With arbitrarily high probability, the selected radius stays bounded and no point far away in the other coordinate enters either marginal neighborhood (lemmas A.4 and A.5). Within the remaining bounded region, the unique boundary neighbor and absence of distance ties make the strict counts stable under small perturbations (lemma A.6). Lemma A.7 combines these steps to establish convergence of the marginal counts to those of the process with intensity equation (9).

Scale that process by $h = \{u\varphi_p(a)\}^{1/p}$. It has exactly the unit-marginal marked representation equation (10), so proposition A.9 applies. Its Gamma arrival determines the joint radius through equation (12); its two marginal-only counts are independent $\mathrm{Pois}(\mu_u)$ variables conditional on the query values and G_k. The first $k - 1$ joint neighbors enter both marginal balls, while the boundary neighbor enters exactly one. Its independent fair orientation follows from the sign-swap $(s', t') \mapsto (-t', -s')$, which preserves the conditional intensity and exchanges the boundary marginals. This gives equation (14). The stopped-process and random-strip arguments establishing these conditional independence statements are proved in lemma A.8 and proposition A.9.

For a bounded continuous test function of the query values and counts, the conditional expectation now converges at each fixed (a, b). Integrating against the fixed Gaussian query law by bounded convergence gives the joint convergence with (A, B). Finally, saturation in equation (14) is exactly $P_x = P_y = 0$. Conditional independence gives its probability $e^{-2\mu_u}$, proving equation (15). □

4 Estimator Bias

The limiting counts are random, but their expected KSG score determines the estimator's limiting bias:

$$B_{p,k}(u) = \log u + \psi(k) - \mathbb{E}[\psi(M_x + 1) + \psi(M_y + 1)]. \tag{16}$$

For computation, define the Poisson-averaged score, for integers $m \geq 1$,

$$g_m(\mu) = \mathbb{E}[\psi\{m + \mathrm{Pois}(\mu)\}]. \tag{17}$$

Either orientation in equation (14) has conditional expected score sum $g_k(\mu_u) + g_{k+1}(\mu_u)$. Hence the equivalent computational representation is

$$B_{p,k}(u) = \log u + \psi(k) - \mathbb{E}[g_k(\mu_u) + g_{k+1}(\mu_u)],$$

where this outer expectation is over (A, B, G_k). To compute the bias, draw A, B, G_k, solve one scalar equation for R, and evaluate the one-dimensional overlap integrals. There is no need to simulate the entire Poisson process.

Theorem 4.1 (Critical estimator bias)**.** *Under equations (1), (3) and (4), with fixed p, k, if $u_n \to u \in (0, \infty)$, then*

$$\widehat{I}_{n,k} - I_n \longrightarrow B_{p,k}(u) \text{ in } L^2. \tag{18}$$

Proof. We first show that the mean error converges to $B_{p,k}(u)$, then show that the estimator's variance tends to zero.

Limiting mean. For observation i, write

$$\ell_{i,n} = \psi(k) - \psi(N_{x,i} + 1) - \psi(N_{y,i} + 1),$$

and let $\ell^{(u)} = \psi(k) - \psi(M_x + 1) - \psi(M_y + 1)$. The channel has $I_n = \log n - \log u_n$, so the exact estimator formula gives

$$\widehat{I}_{n,k} - I_n = \psi(n) - \log n + \log u_n + \frac{1}{n}\sum_{i=1}^{n} \ell_{i,n}. \tag{19}$$

Proposition 3.1 gives convergence in distribution of the counts in a typical neighborhood. The finite-sample moment bound lemma A.11 supplies, for every $0 < \alpha < 1$,

$$\sup_{n>k} \mathbb{E}\{(N_{x,1} + 1)^\alpha + (N_{y,1} + 1)^\alpha\} < \infty.$$

For every finite q, $|\psi(m + 1)|^q \le C_{q,\alpha}\{1 + (m + 1)^\alpha\}$ for $m \ge 0$. Choosing $q > 2$ makes the neighborhood scores and their squares uniformly integrable. The discrete count mapping and this bound give the moment convergence in lemma C.5, in particular $\mathbb{E}\ell_{1,n} \to \mathbb{E}\ell^{(u)}$. By exchangeability,

$$\mathbb{E}\left[\frac{1}{n}\sum_i \ell_{i,n}\right] = \mathbb{E}\ell_{1,n} \longrightarrow \mathbb{E}\ell^{(u)}.$$

Since $\psi(n) - \log n \to 0$ and $\log u_n \to \log u$, equations (16) and (19) imply $\mathbb{E}(\widehat{I}_{n,k} - I_n) \to B_{p,k}(u)$.

Variance. Neighborhood scores share observations, so their variance requires a separate argument. The replacement bound theorem C.3 gives

$$\mathrm{Var}(\widehat{I}_{n,k}) \le C_{p,k}\frac{(1 + \log n)^2}{n} \longrightarrow 0.$$

Replacing one observation changes only a bounded number of joint kth radii, but can change many more marginal counts. For a unit marginal-count change, the digamma score changes by $1/m$, where m is the larger count. Summing these changes by marginal rank using lemma C.2 gives total score change $O_{p,k}(\log n)$. Efron–Stein then yields the displayed variance bound, uniformly in the positive noise level.

Finally,

$$\mathbb{E}|\widehat{I}_{n,k} - I_n - B_{p,k}(u)|^2 = \mathrm{Var}(\widehat{I}_{n,k}) + \{\mathbb{E}(\widehat{I}_{n,k} - I_n) - B_{p,k}(u)\}^2 \longrightarrow 0.$$

This proves equation (18). □

Remark 4.2 (Additive error and sampling fluctuations). The theorem concerns a fixed critical limit $u \in (0, \infty)$, with p, k fixed and scores measured in nats. In particular,

$$\mathbb{E}[\widehat{I}_{n,k} - I_n] \longrightarrow B_{p,k}(u),$$
$$\mathbb{E}[(\widehat{I}_{n,k} - I_n)^2] \longrightarrow B_{p,k}(u)^2.$$

Thus $B_{p,k}(u)$ is the limiting bias, and its square is the limiting mean-squared error. Although $I_n = \log n - \log u_n$ diverges, its additive estimation error can persist after sampling fluctuations vanish. A relative-error plot can therefore conceal the bias. The variance order above is known [4, 13]; the exact-convention proof makes its validity at every positive noise level explicit.

The same law gives the limiting saturation probability $q_{p,k}(u) = \mathbb{E}e^{-2\mu_u}$. In a fixed critical regime, $\overline{S}_{n,k} \to q_{p,k}(u)$ in L^2, as proved with the curve's continuity, strict decrease, and endpoints in corollary 6.1. Saturation records whether both extra marginal-only counts are zero; the bias depends on the full count distribution. A low saturation fraction need not imply a small error.

5 Bias Endpoints and Count Regimes

We first study $B_{p,k}(u)$ as $u \to 0$ and $u \to \infty$. Separately, we establish count limits for finite Gaussian samples along sequences with $u_n \to 0$ or $u_n \to \infty$.

5.1 Endpoints of the limiting bias

Theorem 5.1 (Endpoints of the critical bias)**.** *For every fixed $p, k \geq 1$,*

$$B_{p,k}(u) = \log u - \psi(k+1) + O_{p,k}(u^{1/p}), \qquad u \downarrow 0, \tag{20}$$
$$B_{p,k}(u) \longrightarrow 0, \qquad u \uparrow \infty. \tag{21}$$

Proof. Use the common (A, B, G) construction in proposition 3.1, with $G \sim \text{Gamma}(k, 1)$, and write $\mu = \mu_u$. By equations (16) and (17),

$$B_{p,k}(u) = \log u + \psi(k) - \mathbb{E}\{g_k(\mu) + g_{k+1}(\mu)\}.$$

For the thin endpoint, condition on μ and let $P \sim \text{Pois}(\mu)$. The digamma recurrence gives, for every integer $m \geq 1$,

$$0 \leq \psi(m+P) - \psi(m) = \sum_{r=0}^{P-1} \frac{1}{m+r} \leq \frac{P}{m}. \tag{22}$$

Thus $0 \leq g_m(\mu) - \psi(m) \leq \mu/m$, and

$$-\left(\frac{1}{k} + \frac{1}{k+1}\right)\mathbb{E}\mu \leq B_{p,k}(u) - \{\log u - \psi(k+1)\} \leq 0.$$

The bound $\mathbb{E}\mu \leq C_{p,k}u^{1/p}$ from lemma D.1 proves equation (20), including $p = 1$.

For the regular endpoint, lemma D.2 gives

$$\frac{\mu}{\sqrt{u}} \longrightarrow \left\{\frac{G\varphi_p(A)}{\varphi_p(B)}\right\}^{1/2} \text{ almost surely.}$$

The logarithmic convergence in lemma D.3 then gives

$$2\mathbb{E}\log\mu - \log u \longrightarrow \mathbb{E}\log G + \mathbb{E}\log\varphi_p(A) - \mathbb{E}\log\varphi_p(B) = \psi(k).$$

Here the Gaussian terms cancel because A, B are identically distributed, and the Gamma integral gives $\mathbb{E}\log G = \psi(k)$. To control the difference between the Poisson scores and their logarithms, lemma D.4 gives, for $N \sim \mathrm{Pois}(\mu)$,

$$g_m(\mu) - \log\mu = E_1(\mu) + \sum_{j=1}^{m-1} \mathbb{E}\frac{1}{N+j},\ E_1(\mu) = \int_\mu^\infty \frac{e^{-t}}{t}\,\mathrm{d}t.$$

This nonnegative correction tends to zero as $\mu \to \infty$ and is bounded by $C_m + (-\log\mu)_+$. For $u \geq 1$, set $M = G + \mu$ and $\vartheta = \mu/M$. Then

$$(-\log\mu)_+ \leq \left(-\log\frac{M}{\sqrt{u}}\right)_+ + (-\log\vartheta)_+.$$

The bounds equations (D.14), (D.18) and (D.19) in the proof of lemma D.3 dominate this expression, uniformly for $u \geq 1$, by an integrable multiple of $(1 + \|A\|_2^2 + \|B\|_2^2 + \|B\|_1 + |\log G| + \log(1+G))$. Since $\mu \to \infty$ almost surely, dominated convergence proves

$$\mathbb{E}\{g_k(\mu) + g_{k+1}(\mu) - 2\log\mu\} \longrightarrow 0. \tag{23}$$

Consequently,

$$B_{p,k}(u) = \log u + \psi(k) - 2\mathbb{E}\log\mu + o(1) \longrightarrow 0,$$

which proves equation (21). □

The leading thin-endpoint term is the limiting difference between the ceiling $C_{n,k}$ and the truth I_n. For every finite $u > 0$, $\mu_u > 0$ and strict increase of ψ also give

$$B_{p,k}(u) < \log u - \psi(k+1).$$

The expansion therefore implies negative bias for all sufficiently small u, but this ceiling comparison does not establish a global sign or monotonicity for $B_{p,k}$.

5.2 Thin and Regular Regimes

Theorem 5.2 (Neighborhood count regimes)**.** *For the channel and counts in equations (1) and (3), with fixed p, k:*

(a) *If $u_n \to 0$, then*

$$\mathbb{P}(S_{1,n} = 1) \to 1,\ \overline{S}_{n,k} \to 1 \text{ in } L^1,\ \frac{\widehat{I}_{n,k}}{\log n} \to 1 \text{ in } L^1.$$

(b) *If $u_n \to u \in (0,\infty)$, the query values and count pair converge jointly as in proposition 3.1.*

(c) *If $\delta_n \to 0$ and $u_n \to \infty$, then*

$$\left(\frac{N_{x,1}}{\sqrt{u_n}}, \frac{N_{y,1}}{\sqrt{u_n}}\right) \Rightarrow (L_{p,k}, L_{p,k}),\ L_{p,k} = \left\{G_k \frac{\varphi_p(U)}{\varphi_p(V)}\right\}^{1/2}, \tag{24}$$

where $U, V \sim N(0, I_p)$ and $G_k \sim \mathrm{Gamma}(k,1)$ are mutually independent. Consequently, $\overline{S}_{n,k} \to 0$ in L^1.

Proof. For part (a), the assumption $u_n \to 0$ is exactly that of theorem B.5, which proves all three assertions. Its proof uses the coincidence of the limiting joint and marginal small-ball masses to eliminate marginal-only observations. It then bounds the score deficit of a nonsaturated neighborhood by $O_k(\log n)$ to obtain the normalized estimator limit. Part (b) is proposition 3.1.

For part (c), theorem B.8 applies under the stated assumptions $\delta_n \to 0$ and $u_n \to \infty$. It proves joint convergence of the query values, scaled joint radius, and scaled marginal counts. Taking the count coordinates gives equation (24), including the independent Gaussian and Gamma variables in $L_{p,k}$.

Since $L_{p,k} > 0$ almost surely, the count limit implies divergence of each marginal count in probability: for any fixed M and $\epsilon > 0$, eventually $M/\sqrt{u_n} < \epsilon$, and portmanteau gives

$$\limsup_n \mathbb{P}(N_{x,1} \le M) \le \mathbb{P}(L_{p,k} \le \epsilon).$$

Let $\epsilon \downarrow 0$, and use the same argument for $N_{y,1}$. Saturation forces $N_{x,1} \le k$, so exchangeability yields

$$\mathbb{E}|\overline{S}_{n,k}| = \mathbb{P}(S_{1,n} = 1) \le \mathbb{P}(N_{x,1} \le k) \longrightarrow 0,$$

completing part (c). □

Part (a) concerns a typical neighborhood and the estimator divided by $\log n$. It does not assert that every neighborhood saturates or that the unnormalized gap from the ceiling vanishes. The shrinking-noise assumption in part (c) is essential: fixed positive noise can give different marginal limit constants, as remark B.9 shows.

Proposition 5.3 (Moments of the regular count limit)**.** *The variable $L_{p,k}$ in equation (24) is positive and finite almost surely. For $0 < q < 2$,*

$$\mathbb{E}L_{p,k}^q = \frac{\Gamma(k + q/2)}{\Gamma(k)}(1 - q^2/4)^{-p/2}. \tag{25}$$

For every $q \ge 2$, this moment is infinite. In particular, $\mathbb{E}L_{p,k} < \infty$ but $\mathbb{E}L_{p,k}^2 = \infty$.

Proof. The normalizing constants cancel in the density ratio:

$$\frac{\varphi_p(U)}{\varphi_p(V)} = \exp\left\{\frac{\|V\|_2^2 - \|U\|_2^2}{2}\right\}.$$

Consequently,

$$L_{p,k} = G_k^{1/2} \exp\left\{\frac{\|V\|_2^2 - \|U\|_2^2}{4}\right\}, \tag{26}$$

which is positive and finite almost surely. For $q > 0$, Tonelli's theorem and independence give

$$\mathbb{E}L_{p,k}^q = \mathbb{E}G_k^{q/2}\, \mathbb{E}\exp\{q\|V\|_2^2/4\}\, \mathbb{E}\exp\{-q\|U\|_2^2/4\}. \tag{27}$$

The Gamma moment is

$$\mathbb{E}G_k^{q/2} = \frac{\Gamma(k + q/2)}{\Gamma(k)}.$$

For a standard p-variate Gaussian W,

$$\mathbb{E}e^{t\|W\|_2^2} = (1 - 2t)^{-p/2} \text{ if } t < \frac{1}{2},$$

and the expectation is infinite when $t \geq 1/2$. Thus, for $0 < q < 2$,

$$\mathbb{E}e^{q\|V\|_2^2/4} = (1 - q/2)^{-p/2}, \ \mathbb{E}e^{-q\|U\|_2^2/4} = (1 + q/2)^{-p/2}.$$

Multiplication in equation (27) proves equation (25). If $q \geq 2$, the V-factor is infinite, while the Gamma and U-factors are strictly positive and finite. Tonelli's theorem therefore gives $\mathbb{E}L_{p,k}^q = \infty$. Taking $q = 1$ and $q = 2$ proves the final assertion. □

Corollary 5.4 (Polynomial shrinking-noise paths)**.** *If $\delta_n = n^{-2a/p}$, $a > 0$, then $I_n = a \log n$ and $u_n = n^{1-a}$. For $a > 1$,*

$$\frac{\widehat{I}_{n,k}}{I_n} \longrightarrow \frac{1}{a} \text{ in } L^1.$$

For $a = 1$, $\widehat{I}_{n,k} - I_n \to B_{p,k}(1)$ in L^2. For $0 < a < 1$, the regular count law equation (24) applies.

Proof. If $a > 1$, then $u_n = n^{1-a} \to 0$. By theorem 5.2, $\widehat{I}_{n,k}/\log n \to 1$ in L^1, so

$$\frac{\widehat{I}_{n,k}}{I_n} = \frac{1}{a}\frac{\widehat{I}_{n,k}}{\log n} \longrightarrow \frac{1}{a} \text{ in } L^1.$$

If $a = 1$, then $u_n = 1$, and theorem 4.1 gives the stated L^2 bias limit. If $0 < a < 1$, then $u_n = n^{1-a} \to \infty$ and $\delta_n = n^{-2a/p} \to 0$, exactly the two assumptions of theorem 5.2 part (c). □

Remark 5.5 (Count limits and score limits). The infinite second moment of $L_{p,k}$ concerns the regular count coefficient; it does not contradict the L^2 convergence of the logarithm-like estimator score in a fixed critical window. Neither the regular count law nor the endpoint $B_{p,k}(u) \to 0$ proves estimator consistency along every sequence with $u_n \to \infty$. The endpoint takes $n \to \infty$ at fixed u before increasing u; interchanging these limits would require uniform control of the centered score, not just the count scaling.

6 Saturation-Based Bias Correction

Within the Gaussian model, the limiting saturation fraction determines u. We estimate u from the observed saturation fraction, then subtract the predicted bias at that estimate. The correction is consistent on a fixed critical interval.

6.1 The saturation curve

Corollary 6.1 (Critical saturation curve)**.** *Let $q_{p,k}(u) = \mathbb{E}e^{-2\mu_u}$. This function is continuous and strictly decreasing on $(0, \infty)$, takes values in $(0, 1)$, and satisfies*

$$1 - q_{p,k}(u) = O_{p,k}(u^{1/p}), \qquad u \downarrow 0, \tag{28}$$

$$q_{p,k}(u) \longrightarrow 0, \qquad u \uparrow \infty. \tag{29}$$

If $u_n \to u \in (0, \infty)$, then

$$\overline{S}_{n,k} \longrightarrow q_{p,k}(u) \text{ in } L^2. \tag{30}$$

Proof. Use one common realization of mutually independent $A, B \sim N(0, I_p)$ and $G_k \sim \text{Gamma}(k, 1)$ for all $u > 0$. Put $h_u = \{u\varphi_p(A)\}^{1/p}$, let R_u solve $\Lambda_{h_u,B}(R_u) = G_k$, and set $\mu_u = (2R_u)^p - G_k$.

Fix finite query values (a, b), a positive arrival G_k, and $r > 0$. For $0 < h_1 < h_2$, each difference

$$(2r - h_1|W - b_j|)_+ - (2r - h_2|W - b_j|)_+$$

is nonnegative and is strictly positive on $0 < |W - b_j| < 2r/h_2$, an event of positive Gaussian probability. Every overlap factor is positive and strictly decreasing in h, so $\Lambda_{h_2,b}(r) < \Lambda_{h_1,b}(r)$. Since the radial cumulative mean is continuous and strictly increasing from zero to infinity, its root at G_k is strictly increasing in h. Thus R_u, and hence μ_u, is strictly increasing in u.

The overlap factors are jointly continuous in (h, r), by dominated convergence on compact sets. To check root continuity at h_0, let R_0 be its root and choose $0 < \epsilon < R_0$. Strict radial increase puts G_k strictly between $\Lambda_{h_0,b}(R_0 - \epsilon)$ and $\Lambda_{h_0,b}(R_0 + \epsilon)$. Joint continuity preserves these two inequalities for h near h_0, trapping the root in $(R_0 - \epsilon, R_0 + \epsilon)$. Consequently R_u and μ_u have continuous paths.

Because $0 < \mu_u < \infty$ almost surely, $e^{-2\mu_u} \in (0, 1)$. Bounded convergence gives continuity of $q_{p,k}$. If $u_1 < u_2$, then $e^{-2\mu_{u_1}} - e^{-2\mu_{u_2}} > 0$ almost surely; its expectation is strictly positive. Hence $q_{p,k}$ is strictly decreasing and lies in $(0, 1)$. For $0 < u \le 1$, the small-strip-mass bound lemma D.1 gives

$$0 \le 1 - q_{p,k}(u) = \mathbb{E}(1 - e^{-2\mu_u}) \le 2\mathbb{E}\mu_u \le C_{p,k}u^{1/p},$$

proving equation (28) and the limit one at zero. The large-u overlap scaling lemma D.2 gives

$$\frac{\mu_u}{\sqrt{u}} \longrightarrow \left\{G_k \frac{\varphi_p(A)}{\varphi_p(B)}\right\}^{1/2} > 0 \text{ almost surely.}$$

Thus $e^{-2\mu_u} \to 0$, and bounded convergence proves equation (29).

For the empirical limit, the stopped-process law proposition 3.1 and exchangeability give

$$\mathbb{E}\overline{S}_{n,k} = \mathbb{P}(S_{1,n} = 1) \longrightarrow q_{p,k}(u).$$

The saturation event is a finite set in the discrete count space, so its probability converges with the count law. Shared observations do not prevent concentration: lemma E.1 gives

$$\text{Var}(\overline{S}_{n,k}) \le \frac{D_{p,k}^2}{2n}, \; D_{p,k} = 1 + 2\kappa_{2p}k + 4\kappa_p(k + 1).$$

The fixed-rank mechanism is useful here. After one observation is replaced, at most $2\kappa_{2p}k$ common query radii change. At an unchanged radius, an indicator can change only if one configuration is saturated. Its marginal counts are at most k, and the other configuration's counts are at most $k + 1$. The old or new point responsible for a change therefore has marginal rank at most $k + 1$. The reverse-neighbor bound in each marginal space controls these queries by $4\kappa_p(k+1)$; adding the replaced query gives $D_{p,k}$. Efron–Stein yields the variance inequality without assuming independent neighborhood scores. Finally,

$$\mathbb{E}|\overline{S}_{n,k} - q_{p,k}(u)|^2 = \text{Var}(\overline{S}_{n,k}) + \{\mathbb{E}\overline{S}_{n,k} - q_{p,k}(u)\}^2 \longrightarrow 0,$$

which proves equation (30). □

6.2 Bias Correction on a Fixed Interval

Continuity, strict decrease, and the endpoint limits make $q_{p,k}$ a bijection from $(0,\infty)$ onto $(0,1)$. On a fixed transition window, its inverse turns the observed saturation fraction into an estimate of u. The following result uses the same sample for that estimate and for the original KSG score.

Corollary 6.2 (Calibration on a fixed transition window)**.** *Fix p, k, and choose $K = [a, b]$ with $0 < a < b < \infty$. Suppose the channel and estimator satisfy equations (1), (3) and (4) and $u_n \to u \in K$. Write $q = q_{p,k}$, $B = B_{p,k}$, and let $\Pi_{[c,d]}(s) = \min\{d, \max\{c, s\}\}$ denote clipping. The function B is continuous on $(0,\infty)$. Define*

$$\widehat{u}_n = q^{-1}\big(\Pi_{[q(b),q(a)]}(\overline{S}_{n,k})\big)\,,\ \widehat{I}^{\mathrm{cal}}_{n,k} = \widehat{I}_{n,k} - B(\widehat{u}_n). \tag{31}$$

Then

$$\widehat{u}_n \longrightarrow u,\ \widehat{I}^{\mathrm{cal}}_{n,k} - I_n \longrightarrow 0 \text{ in } L^2. \tag{32}$$

Proof. Lemma E.2 proves that $B_{p,k}$ is finite and continuous. It is therefore bounded and uniformly continuous on the fixed interval K. We use these properties below to show that estimating u consistently also gives a consistent bias correction.

Let $J = [q(b), q(a)]$. By corollary 6.1, the restriction of q to K is a continuous, strictly decreasing bijection onto J. Its inverse is continuous: if $s_j \to s$ in J, every convergent subsequence of the bounded sequence $q^{-1}(s_j)$ has a limit v satisfying $q(v) = s$. Injectivity forces $v = q^{-1}(s)$, and compactness gives convergence of the whole sequence. Therefore

$$F(s) = q^{-1}(\Pi_J(s)),\ s \in [0,1],$$

is continuous and bounded in K, with $F(q(u)) = u$, including when $u = a$ or $u = b$.

Given $\epsilon > 0$, choose $\eta > 0$ such that $|F(s) - u| < \epsilon$ whenever $|s - q(u)| < \eta$. The empirical mean-square limit equation (30) and Markov's inequality give

$$\begin{aligned}\mathbb{E}|\widehat{u}_n - u|^2 &\le \epsilon^2 + (b-a)^2\mathbb{P}\{|\overline{S}_{n,k} - q(u)| \ge \eta\}\\ &\le \epsilon^2 + \frac{(b-a)^2}{\eta^2}\mathbb{E}|\overline{S}_{n,k} - q(u)|^2.\end{aligned}$$

Hence $\limsup_n \mathbb{E}|\widehat{u}_n - u|^2 \le \epsilon^2$; letting $\epsilon \downarrow 0$ proves $\widehat{u}_n \to u$ in L^2.

For the bias subtraction, let $M = \sup_{v\in K}|B(v)| < \infty$. Uniform continuity gives, for any $\epsilon > 0$, an $\eta > 0$ such that $|B(v) - B(u)| < \epsilon$ when $|v - u| < \eta$. Then

$$\mathbb{E}|B(\widehat{u}_n) - B(u)|^2 \le \epsilon^2 + 4M^2\mathbb{P}\{|\widehat{u}_n - u| \ge \eta\}$$

Thus $\limsup_n \mathbb{E}|B(\widehat{u}_n) - B(u)|^2 \le \epsilon^2$. Letting $\epsilon \downarrow 0$ proves $B(\widehat{u}_n) \to B(u)$ in L^2. Finally,

$$\begin{aligned}\mathbb{E}|\widehat{I}^{\mathrm{cal}}_{n,k} - I_n|^2 &\le 2\mathbb{E}|\widehat{I}_{n,k} - I_n - B(u)|^2 + 2\mathbb{E}|B(\widehat{u}_n) - B(u)|^2\\ &\longrightarrow 0\end{aligned}$$

by theorem 4.1. No independence between the estimated bias and the original KSG score is used. The argument needs only $u_n \to u \in K$, not that every u_n belongs to K. □

Remark 6.3 (Scope of bias correction). The interval is fixed before seeing the sample. Clipping keeps the estimated u within this interval, but does not guarantee an accurate correction at finite sample sizes. Estimation can be difficult when saturation is rare, and errors in the computed curves can affect the correction. The result concerns this Gaussian model on a fixed critical interval. Numerical lookup tables require approximation control, such as the uniform conditions in proposition E.3.

Table 1: Finite-sample bias and neighborhood saturation at $n = 8000$, $k = 5$, $u = 10$. Intervals are 95% Student-t intervals for setting means across 30 independent repetitions; saturation is expressed as a percentage.

Dimension	Mean bias in bits [95% interval]	Saturation [95% interval]
1	-0.406 $[-0.412, -0.401]$	4.737% [4.635, 4.839]
2	-1.047 $[-1.054, -1.041]$	1.222% [1.165, 1.278]
4	-2.398 $[-2.404, -2.391]$	0.0271% [0.0193, 0.0348]

7 Numerical Results

We test whether the Poisson law predicts KSG's bias and saturation probability in finite Gaussian samples. We then examine concentration around the predicted bias, test saturation-based bias correction, and evaluate the large-u endpoint of the limiting law. The exact Gaussian information lets us measure estimation error directly.

7.1 Bias and Saturation

We use $n \in \{250, 500, 1000, 2000, 4000, 8000\}$, $p \in \{1, 2, 4\}$, and $k \in \{3, 5, 10\}$. For fixed $u < n$, we set $\delta_n = (u/n)^{2/p}$. As n increases, this reduces the channel noise and raises the true information. The fixed-u comparisons therefore test approach to the predicted bias along changing channels, rather than estimation of one fixed distribution. Each setting has 30 independent whole-dataset repetitions. Within a repetition, source and noise draws are reused across channel settings and neighbor indices for paired comparisons. Appendix F gives the simulation methods and numerical checks.

The limiting study uses 30,000 independent Monte Carlo draws per reference setting. Each draw consists of Gaussian source and noise values A, B, a Gamma arrival, and the conditional marginal-only counts. Reusing these draws across u pairs the calculations. A separate direct simulation of the marked Poisson process at an asymmetric query checks the Gamma/strip representation and the equally likely boundary orientations. Figure 2 tests the transition across u: the top panels show saturation probability, and the bottom panels show mean KSG error relative to the true information, in bits. The curves are independent Poisson predictions, not fits to the finite-sample means shown by open points at $n = 8000$. Columns separate dimensions; colors distinguish neighbor index. The study also retains all smaller sample sizes.

Across 114 settings at $n = 8000$, the root-mean-square differences between measured means and Poisson predictions are 0.00238 bit for bias and 0.3215 percentage points for saturation. The theory therefore accurately predicts KSG's mean error over the tested grid. Saturation predictions are also close in absolute terms, although some discrepancies exceed their Monte Carlo uncertainty.

7.2 Bias Below the Ceiling

The ceiling bounds what KSG can report; it does not guarantee accuracy below that bound. At $n = 8000$, $k = 5$, and $u = 10$, the true information is 9.644 bits in every dimension, approximately 0.860 bit below the general-position ceiling. Yet KSG underestimates it by about 2.4 bits on average in dimension four, while almost no neighborhoods saturate (Table 1). The dimension-four setting is an additional tested point, outside that dimension's original plotted reference grid.

The three dimensions have the same ceiling gap but different mean errors. Saturation alone

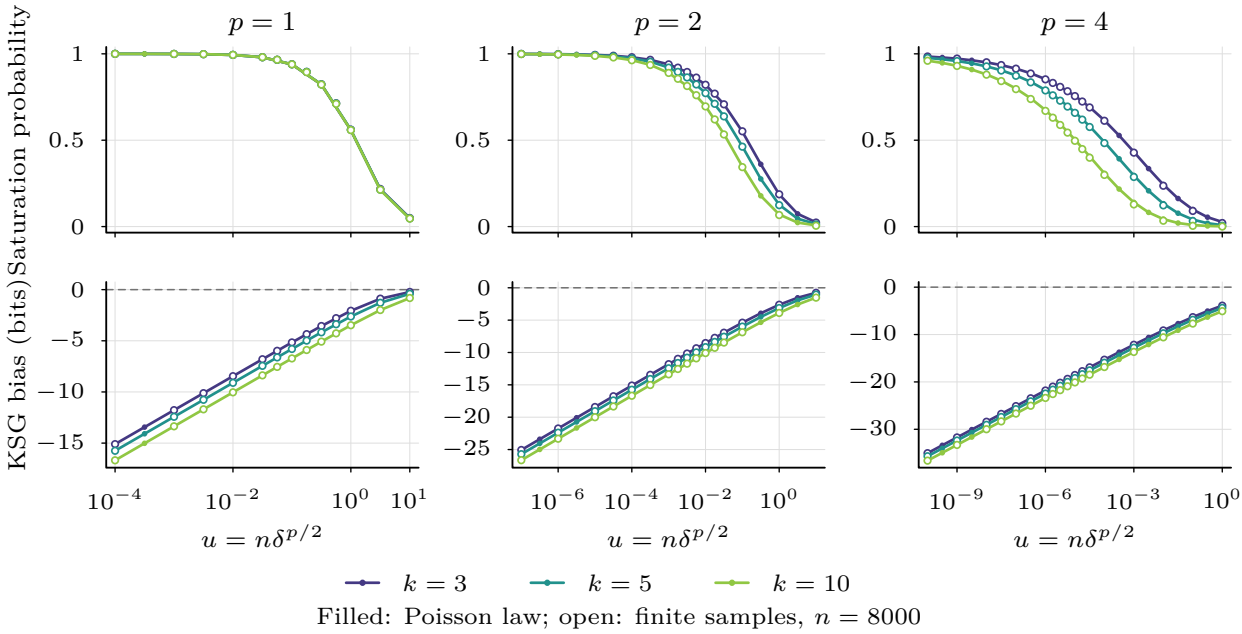


Figure 2: Critical saturation probability (top) and additive bias (bottom). Columns separate dimensions $p = 1, 2, 4$; colors distinguish $k = 3, 5, 10$. Open finite-sample points are means at $n = 8000$, with 95% Student-t intervals across 30 independent repetitions per setting. Poisson curves have pointwise Monte Carlo bands of mean plus or minus 1.96 standard errors. Each dimension uses the u-range shown; the dimension-four extension is shown separately in Figure 5.

therefore misses substantial estimation error. The distinction is already present in the theory: $q_{p,k}$ measures the probability of no extra marginal-only neighbors, whereas $B_{p,k}$ depends on the full distribution of neighborhood scores.

7.3 Convergence at Fixed u

We next test whether KSG's error approaches the predicted bias and its dataset-to-dataset variability decreases as n grows at fixed u. As explained above, the channel changes with n in this experiment. We reuse the simulations in Section 7.1 at $k = 5$, $u \in \{0.01, 0.1, 1\}$, and all six sample sizes; no new finite-sample simulations are used.

Figure 3 subtracts $B_{p,5}(u)/\log 2$ from observed mean bias in its left panels and $q_{p,5}(u)$ from observed mean saturation in its right panels. Zero in the bias panels means accurate prediction of KSG's error, not unbiased estimation. Each point combines uncertainty across 30 independent whole datasets with the 30,000-draw Poisson integration uncertainty. The Poisson calculation is independent of the finite samples, but the same numerical reference is reused at every n within a panel. Its error is therefore shared, not reduced by the number of plotted points.

At fixed $u = 1$, the measured mean bias approaches the Poisson prediction as n increases, although not monotonically. In dimension four at $n = 8000$, KSG still underestimates the information by about 4.38 bits. The theory accurately predicts this persistent error.

Saturation predictions are less accurate in that dimension: at the same setting, the observed saturation fraction is about half the predicted value. Accurate mean-bias predictions can therefore coexist with appreciable relative error for rare saturation events. The uncertainty calculations appear in Appendix F.4.

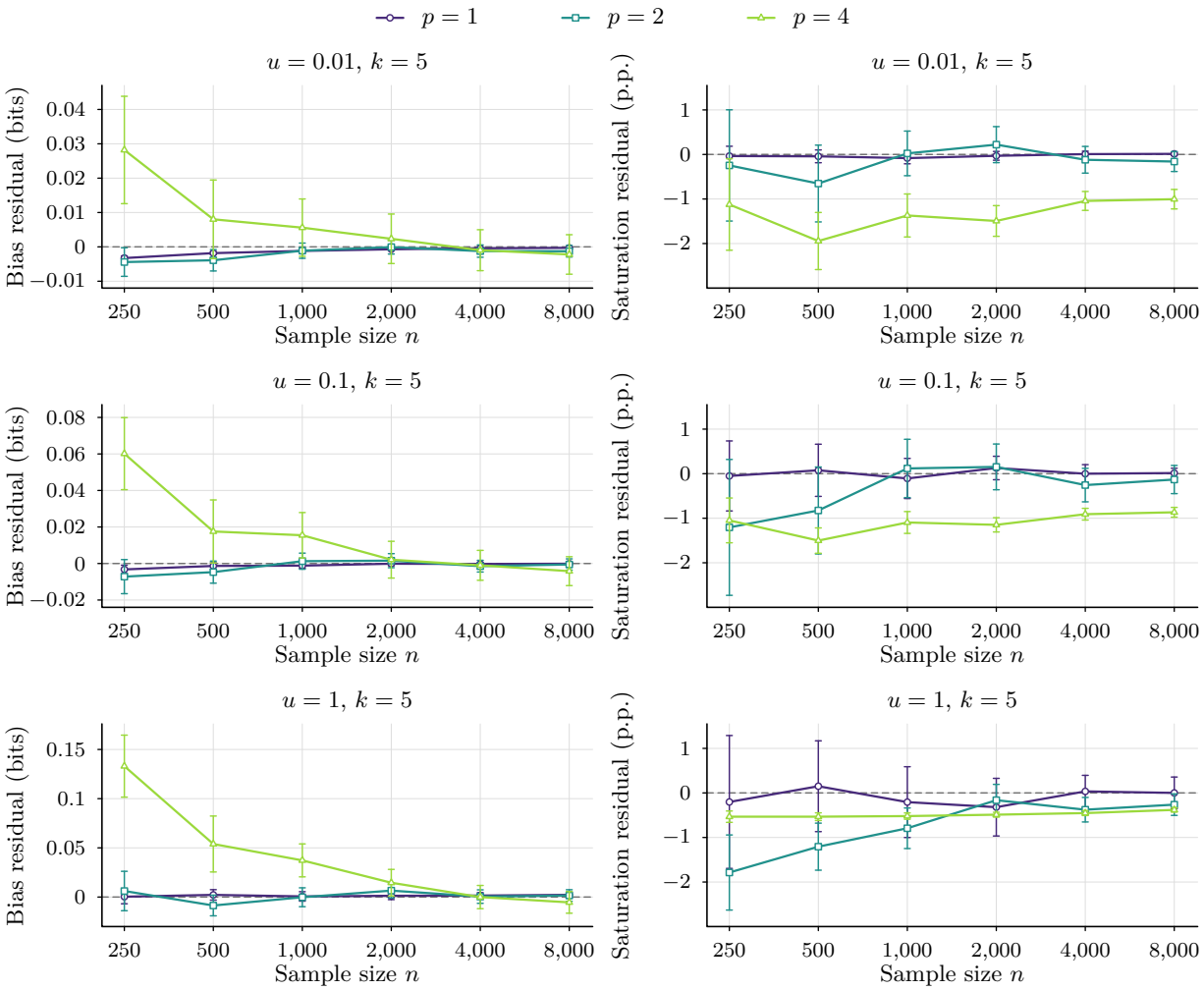


Figure 3: Fixed-u residuals from the simulation study in Section 7.1, with $k = 5$. Rows use $u = 0.01, 0.1, 1$; colors distinguish $p = 1, 2, 4$. Left: mean KSG bias minus the Poisson bias, in bits. Right: mean saturated-neighborhood fraction minus the Poisson probability, in percentage points (100 times the probability difference). Zero is the asymptotic target. Approximate pointwise 95% Welch–Satterthwaite intervals combine independent finite-dataset and Poisson-integration variances; the latter is shared across sample sizes. Numerical quadrature and truncation errors are not included. No convergence rate is fitted.

7.3.1 Mean-Square Convergence

Agreement of the mean error with its predicted limit does not show whether individual estimates concentrate around that limit. We therefore measure the mean-square discrepancy across independent datasets. Write $T_r = (\widehat{I}_{n,5}^{(r)} - I_n)/\log 2$ and $b = B_{p,5}(u)/\log 2$, both in bits. For the independent numerical reference $\widetilde{b}$, let $\bar{d} = \bar{T} - \widetilde{b}$ and let s_T^2 be the unbiased sample variance across the $R = 30$ complete datasets. The empirical mean-square discrepancy decomposes exactly as

$$\widetilde{M}_n = \frac{1}{R}\sum_{r=1}^{R}(T_r - \widetilde{b})^2 = \bar{d}^{\,2} + \frac{R-1}{R}s_T^2. \tag{33}$$

Here $\sqrt{\widetilde{M}_n}$ measures approach to the limiting bias, whereas s_T measures dataset-to-dataset spread without Poisson-reference error. Neither is the raw KSG RMSE relative to the true information.

If $\widetilde{b}$ is an independent unbiased Monte Carlo mean with variance v_b, then

$$\mathbb{E}\widetilde{M}_n = \mathbb{E}(T_1 - b)^2 + v_b.$$

Thus $\widetilde{M}_n - s_b^2$ estimates the mean-squared discrepancy from the limiting bias, and $\bar{d}^{\,2} - s_T^2/R - s_b^2$ estimates its squared-bias component, where s_b is the reference standard error. The latter estimate can be negative from sampling noise; it is neither truncated nor interpreted as a negative population squared bias. These corrections apply to squared errors, not their square roots. Reference error is shared across all six sample sizes and is not divided by the number of datasets or points. Subtracting its estimated variance does not remove quadrature, root-solving, or truncation bias.

Both the root-mean-square discrepancy and dataset SD decrease from $n = 250$ to $n = 8000$ in all nine settings in Table 2. The SD reduction does not depend on the numerical reference: KSG becomes more reproducible while retaining the approximately -4.379-bit mean error reported above. At $u = 1, n = 8000$, subtracting reference variance changes the root-mean-square discrepancies for $p = 1, 2, 4$ from $0.00427, 0.00710, 0.01309$ to $0.00408, 0.00656, 0.01203$ bit. The declining dataset spread supplies concentration evidence; the reference uncertainty identifies the numerical floor of the mean-square comparison.

Table 2: Mean-square convergence diagnostic from the same 30 independent datasets per setting, at $k = 5$. All entries are in bits. RMSE is $\sqrt{\widetilde{M}_n}$, before subtracting reference variance; dataset SD is s_T. The last column reports the standard error of the independent 30,000-draw Poisson reference, shared across n. The table displays the first and last sample sizes; the analysis uses all six. These are descriptive estimates, not confidence intervals or fitted convergence rates.

		RMSE about $\widetilde{b}$		Dataset SD		Reference
p	u	$n = 250$	$n = 8000$	$n = 250$	$n = 8000$	SE
1	0.01	0.00362	0.00040	0.00160	0.00027	0.00011
1	0.1	0.00662	0.00075	0.00587	0.00075	0.00038
1	1	0.01784	0.00427	0.01814	0.00371	0.00125
2	0.01	0.01114	0.00216	0.01040	0.00177	0.00079
2	0.1	0.02461	0.00335	0.02394	0.00337	0.00144
2	1	0.05136	0.00710	0.05187	0.00705	0.00273
4	0.01	0.04819	0.00742	0.03975	0.00721	0.00262
4	0.1	0.07740	0.01001	0.04953	0.00924	0.00366
4	1	0.15472	0.01309	0.08033	0.01212	0.00516

7.4 Bias Correction

We test whether estimating u from saturation and subtracting the predicted bias improves finite-sample accuracy. Corollary 6.2 establishes consistency within a fixed Gaussian transition window. Here we retain every dataset, including those with no saturated neighborhoods or a saturation fraction outside the tabulated range.

This separate study uses $k = 5$, $p = 1, 2, 4$, $n = 1000, 4000, 8000$, and 16 independent source/noise repetitions per setting. These draws are paired across u and independent of the main transition study. Primary and validation Poisson lookups each use 30,000 independent Gaussian/Gamma draws per dimension, with independent seeds. The windows, fixed before lookup simulation, are

$$p = 1 : \ [10^{-1}, 10], \ p = 2 : \ [10^{-2.5}, 10], \ p = 4 : \ [10^{-4.5}, 10^{0.5}].$$

Quarter-decade nodes bracket the finite tests, all off-grid midpoints. We interpolate linearly in $\log u$ to estimate u and evaluate its bias. An observed saturation fraction outside the tabulated range is moved to its nearest endpoint before inversion. Every headline result uses the primary curve; the independent curve measures seed sensitivity.

We compare raw KSG, correction using $B(\widehat{u}_n)$, correction using the *true* u, and the Gaussian estimator based directly on the variance of $X - Y$ in equation (E.4). The true-u correction, labeled "Oracle," uses the same numerical bias table. It is a diagnostic, not an estimator available when u is unknown. The Gaussian estimator exploits the assumed channel model. We record mean error and root-mean-square error (RMSE) per setting, paired squared-error differences, clipping and zero-saturation frequencies, and expected and observed numbers of saturated neighborhoods. RMSE is the square root of the average squared information-estimation error across datasets.

Figure 4 and Table 3 show a large reduction in error relative to raw KSG, though the Gaussian estimator has lower RMSE than the saturation-based correction. The pooled summaries use different windows and do not provide a controlled comparison across dimensions. At the common setting $n = 8000, u \approx 0.7499$, correction again reduces error in all three dimensions, but leaves much larger error in dimension four (Table 4).

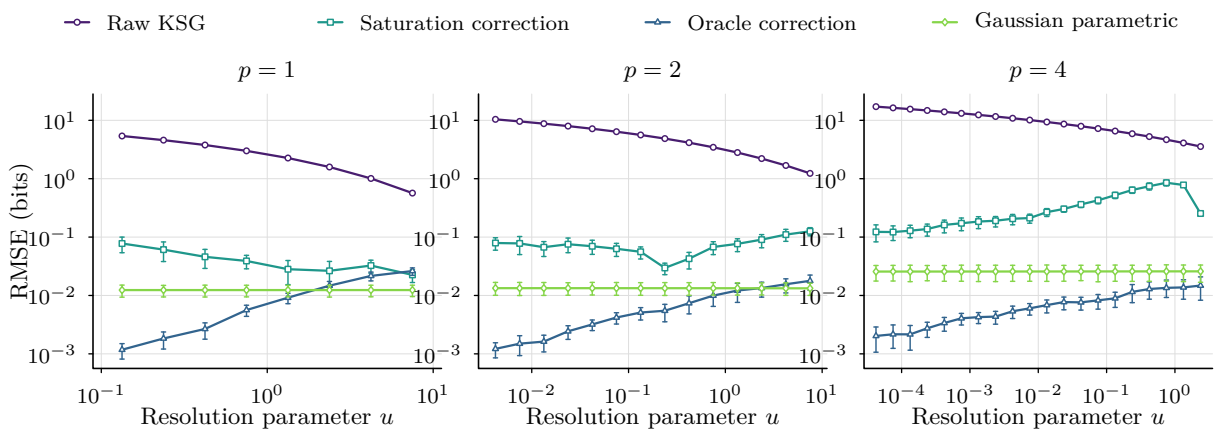


Figure 4: Bias-correction test at $n = 8000, k = 5$, retaining every trial. Each setting has 16 independent datasets; the horizontal settings reuse the same source/noise pools. Error bars are pointwise percentile bootstrap intervals for RMSE from 3,000 resamples of those 16 datasets, with shared resample indices across methods. They condition on the primary lookup; independent-seed sensitivity is evaluated separately. Oracle correction uses the true u, whereas saturation-based correction uses only the observed fraction and a prespecified Gaussian lookup. The parametric curve uses the exact mean correction in equation (E.4). The rightmost dimension-four drop is affected by clipping to the known window; it does not demonstrate more accurate estimation of u.

To identify the source of the remaining dimension-four error, we apply the same bias correction using the true u. Its RMSE falls to 0.01350 bit, compared with 0.02587 bit for the Gaussian estimator. The predicted bias is therefore accurate here when evaluated at the correct parameter. With saturation-based correction, the mean error is -0.8317 bit, with a 95% interval $[-0.9347, -0.7287]$. The lookup predicts about 84 saturated neighborhoods per dataset, against an observed mean of about 44. This leads the procedure to overestimate u and apply too little correction. No estimate in this setting is clipped: estimating u from finite-sample saturation is the bottleneck, rather than clipping or predicting the bias at the correct parameter. At $u \approx 1.3335$ and 2.3714, respectively 13 and 16 of the 16 dimension-four inversions clip at $n = 8000$. The final drop in corrected RMSE therefore does not demonstrate more accurate estimation of u.

Across all 2,016 evaluations, 117 estimates are clipped and 13 datasets have zero saturation; none is excluded. The fixed lookup adds a numerical floor: direct off-grid checks give interpolation discrepancies up to 0.01872 bit for bias and 0.005543 in saturation probability. Replacing it by the independently seeded lookup changes dimension-four corrected estimates by about 0.0387 bit in descriptive RMS at $n = 8000$. These numerical errors lie outside the exact-function conclusion in corollary 6.2; the checks do not certify the uniform convergence assumed in proposition E.3.

Halving the lookup-grid spacing reduces interpolation error but leaves the corrected estimator's mean error nearly unchanged. At $p = 4$, $n = 8000$, and $u \approx 0.75$, it changes from approximately -0.832 to -0.822 bit. The refined lookup still predicts nearly twice the observed number of saturated neighborhoods, and no estimate is clipped. This check uses the same datasets and Monte Carlo draws, rather than a new independent experiment. Coarse interpolation alone therefore does not explain the remaining error.

Saturation-based bias correction removes much of KSG's error within the Gaussian model, but estimating u from saturation limits its accuracy at these sample sizes, especially in dimension four. The Gaussian estimator remains a stronger model-specific benchmark. Non-Gaussian robustness

Table 3: Descriptive RMSE in bits at $k = 5$, pooled over each dimension's off-grid u settings. The windows differ across dimensions. Settings within a pool are dependent, so these pooled summaries have no confidence interval based on independent settings. All trials, including clipped inversions, are retained.

p	n	Raw KSG	Saturation	Oracle	Gaussian
1	1000	3.209	0.1353	0.0157	0.0216
1	4000	3.207	0.0688	0.0131	0.0136
1	8000	3.207	0.0453	0.0137	0.0124
2	1000	6.176	0.1922	0.0185	0.0425
2	4000	6.177	0.1028	0.0095	0.0211
2	8000	6.175	0.0774	0.0090	0.0133
4	1000	10.792	0.6215	0.0288	0.0635
4	4000	10.796	0.4922	0.0121	0.0307
4	8000	10.797	0.4118	0.0082	0.0258

Table 4: Information-estimation RMSE in bits at the matched setting $n = 8000$, $k = 5$, $u \approx 0.7499$, over 16 independent datasets. Unlike Table 3, these entries compare the same u across dimensions.

Dimension p	Raw KSG	Corrected KSG
1	3.000	0.0390
2	3.453	0.0677
4	4.665	0.8525

and one-sided inference are not tested here.

7.5 Large-u Bias in Dimension Four

To examine the large-u endpoint, we extended the Poisson calculation for $p = 4, k = 5$ beyond its original range $u \leq 1$. The initial extension reached 10^8. Its positive bias prompted a further calculation reaching 10^{12}, using two independent pools of 30,000 Monte Carlo draws. We checked numerical sensitivity by increasing the integration order, root-search window, and number of draws. These calculations evaluate the limiting bias function, not finite-sample KSG at those values of u.

To estimate the small bias more precisely, we subtract a control variate with exactly zero expectation:

$$Z = \psi(k) - \log G_k + \frac{\|A\|_2^2 - \|B\|_2^2}{2}, \ \mathbb{E}Z = 0. \tag{34}$$

Subtracting Z from each simulated score preserves the expectation $B_{p,k}(u)$ and reduces Monte Carlo variance. It changes numerical evaluation of the Poisson expectation, not the KSG estimator. Appendix F gives the identity and numerical method.

Pooling 30,000 draws from each of two independent seeds gives the estimates in Table 5. Both seeds separately agree on the signs. The positive region followed by decay toward zero is numerical evidence against assuming an everywhere-negative bias or a monotone approach to zero; neither is a claim of theorem 5.1.

Numerical checks address rare Gaussian source and noise values, unresolved radii, inaccurate overlap integration, and dependence between paired settings. Root residuals, window-tail diagnostics, and independent seeds check resolution. A small root residual certifies the solved numerical

Table 5: Monte Carlo estimates of $B_{4,5}(u)/\log 2$, pooling two independent seeds with 30,000 draws each. Intervals describe pointwise Monte Carlo uncertainty, not finite-sample KSG error.

u	Bias in bits	95% interval
10^4	-0.11341	$[-0.11957, -0.10725]$
10^6	0.03728	$[0.03424, 0.04031]$
10^{12}	0.004254	$[0.004053, 0.004455]$

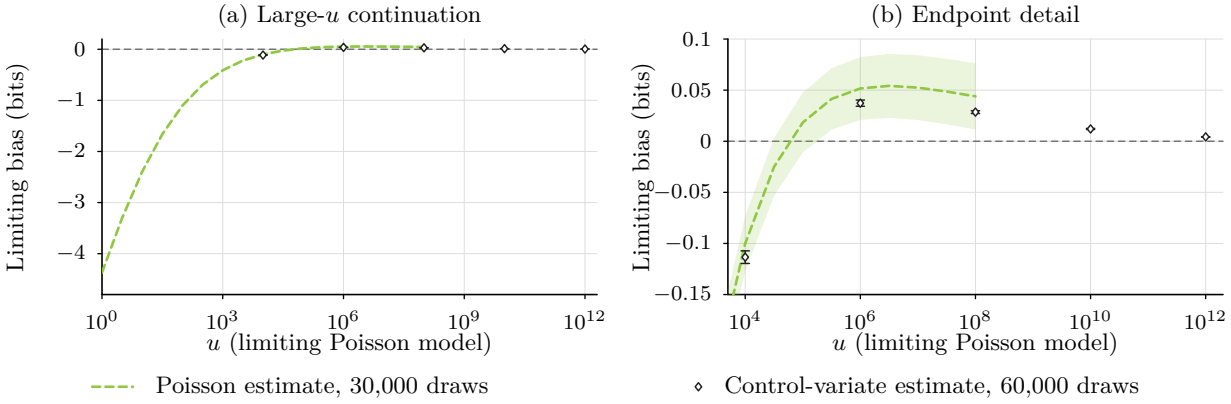


Figure 5: Additional limiting-process calculations for $p = 4, k = 5$. The extended range exposes the approach toward the zero-bias endpoint. The black control-variate points pool two independent seeds, totaling 60,000 draws, and preserve the Poisson expectation exactly. Uncertainty is pointwise Monte Carlo uncertainty, not a statement about finite-sample KSG at these values of u. The sign and shape observed here are numerical properties of the tested Poisson bias function.

equation, not population tail error. The uncertainty bands throughout this section are pointwise; the observed endpoint signs remain numerical evidence.

8 Discussion

The critical Poisson law describes KSG's neighborhood counts and gives mean-square convergence of its estimation error to a computable limiting bias. This quantifies the estimator's behavior at the known exponential-in-information sample scale, including the effects of dimension and neighbor index. A low saturation fraction need not imply small bias, and sampling fluctuations can vanish while an additive error persists.

The saturation curve gives a consistent Gaussian-model correction on a fixed transition interval. In the dimension-four example, the remaining error comes mainly from estimating u: applying the same correction at the true u is accurate even when estimating it from saturation is not. When the Gaussian family is known, the estimator based on the variance of $X - Y$ in proposition E.4 is a more direct estimator.

Remark 8.1 (One-sided interpretation). The ceiling bounds the *estimator*, not its error relative to the true information. The large-u dimension-four calculation supplies numerical evidence of a positive limiting bias; it does not prove its sign. More generally, if $B_{p,k}(u) > 0$ at any particular u, theorem 4.1 implies

$$\mathbb{P}\{\widehat{I}_{n,k} > I_n\} \longrightarrow 1 \text{ whenever } u_n \to u.$$

Indeed, the probability of a nonpositive error is at most $\mathbb{E}|\widehat{I}_{n,k} - I_n - B_{p,k}(u)|^2 / B_{p,k}(u)^2$, which tends to zero. This conditional implication is rigorous; the observed positive sign remains a numerical premise. Thus raw KSG is not justified as an information lower bound by these results. Bias-correction consistency likewise does not imply one-sided coverage.

The scope is fixed p, fixed k, the exact strict-radius convention, and the shrinking Gaussian channel equation (1). Other source and noise laws would require corresponding control of remote marks and score tails, and a replacement for the Gaussian overlap calculation. Consistency along all regular-side sequences would require centered-score control uniform in a growing u. These are extensions of the present explicit count and bias law, not consequences of its fixed-window theorem.

Acknowledgments

We thank Aayush Rajesh and Matthew Ho for their helpful discussions and feedback.
AI statement. OpenAI's ChatGPT and Codex tools assisted initial theoretical exploration, proof drafting and checking (Sections 3–6 and Appendices A–E), experimental code and plotting (Section 7 and Appendix F). The authors are responsible for the final manuscript.

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A Local Poisson Convergence and Marginal Counts

Marginal neighborhoods can include points far away in the other coordinate, so convergence on bounded regions alone does not establish count convergence. We prove the process and count limits in proposition 3.1, including control of these distant points, then establish the fractional-count moment bound needed for theorem 4.1.

A.1 Exact setting and statement

Fix $p, k \geq 1$, assume

$$u_n = n\delta_n^{p/2} \longrightarrow u \in (0, \infty), \tag{A.1}$$

and consider query observation $i = 1$. No monotonicity of δ_n is assumed; equation (A.1) itself implies $\delta_n \to 0$. Let

$$A_n = X_1, \ B_n = Z_1.$$

Conditional on $(A_n, B_n) = (a, b)$, define, for $j = 2, \ldots, n$,

$$\xi_{j,n} = \frac{X_j - a}{\sqrt{\delta_n}}, \ \eta_{j,n} = \frac{Y_{j,n} - Y_{1,n}}{\sqrt{\delta_n}}, \ \Xi_n^{a,b} = \sum_{j=2}^{n} \delta_{(\xi_{j,n}, \eta_{j,n})}. \tag{A.2}$$

For a locally finite point measure ξ on $\mathbb{R}^p \times \mathbb{R}^p$, put

$$C_r = \{(s,t) : \|s\|_\infty \leq r, \ \|t\|_\infty \leq r\}, \tag{A.3}$$

$$\rho_k(\xi) = \inf\{r \geq 0 : \xi(C_r) \geq k\}, \tag{A.4}$$

$$\mathcal{N}_x(\xi) = \xi\{(s,t) : \|s\|_\infty < \rho_k(\xi)\}, \tag{A.5}$$

$$\mathcal{N}_y(\xi) = \xi\{(s,t) : \|t\|_\infty < \rho_k(\xi)\}. \tag{A.6}$$

The sets in equations (A.5) and (A.6) are noncompact.

Theorem A.1 (Critical local process and strict-count limit)**.** *Let $A, B \overset{\text{iid}}{\sim} N(0, I_p)$. Conditional on $(A, B) = (a, b)$, let $\Xi^{a,b}$ be a Poisson point process on $\mathbb{R}^p \times \mathbb{R}^p$ with intensity density*

$$\lambda_{a,b}(s,t) = u\varphi_p(a)\varphi_p(b + t - s). \tag{A.7}$$

Then $\rho_k(\Xi^{A,B})$, $\mathcal{N}_x(\Xi^{A,B})$, and $\mathcal{N}_y(\Xi^{A,B})$ are finite almost surely, and

$$\left(A_n, B_n, \frac{\varepsilon_{1,n}}{\sqrt{\delta_n}}, N_{x,1}, N_{y,1}\right) \Rightarrow \left(A, B, \rho_k(\Xi^{A,B}), \mathcal{N}_x(\Xi^{A,B}), \mathcal{N}_y(\Xi^{A,B})\right). \tag{A.8}$$

For every fixed (a, b), the corresponding regular conditional law given $(A_n, B_n) = (a, b)$ also converges.

A.2 Exact coordinate map and conditional Poisson convergence

Lemma A.2 (Exact conditional density)**.** *Conditional on $(A_n, B_n) = (a, b)$, the points in equation (A.2) are iid and satisfy*

$$\eta_{j,n} = (1 - \delta_n)\xi_{j,n} + \sqrt{1 - \delta_n}\,(Z_j - b). \tag{A.9}$$

Their Lebesgue density is

$$q_n^{a,b}(s,t) = \delta_n^{p/2}(1-\delta_n)^{-p/2}\varphi_p(a+\sqrt{\delta_n}s)\varphi_p\left(b+\frac{t-(1-\delta_n)s}{\sqrt{1-\delta_n}}\right). \tag{A.10}$$

On every compact subset of $\mathbb{R}^{2p}$,

$$(n-1)q_n^{a,b}(s,t) \longrightarrow u\varphi_p(a)\varphi_p(b+t-s) \tag{A.11}$$

uniformly.

Proof. Subtract the query channel equation from the jth channel equation:

$$Y_{j,n} - Y_{1,n} = (1-\delta_n)(X_j - a) + \sqrt{\delta_n(1-\delta_n)}(Z_j - b).$$

Division by $\sqrt{\delta_n}$ proves equation (A.9). The inverse map from (s,t) to the independent Gaussian variables (x,z) is

$$x = a + \sqrt{\delta_n}s,\ \ z = b + \frac{t-(1-\delta_n)s}{\sqrt{1-\delta_n}}.$$

Its absolute Jacobian determinant is $\delta_n^{p/2}(1-\delta_n)^{-p/2}$, which proves equation (A.10). Finally, $(n-1)\delta_n^{p/2} \to u$ and $\delta_n \to 0$. Both Gaussian factors and the scalar Jacobian correction converge uniformly on compact sets, proving equation (A.11). □

The map is also exact for the KSG metric:

$$\frac{\max\{\|X_j - X_1\|_\infty, \|Y_{j,n} - Y_{1,n}\|_\infty\}}{\sqrt{\delta_n}} = \max\{\|\xi_{j,n}\|_\infty, \|\eta_{j,n}\|_\infty\}. \tag{A.12}$$

Thus

$$\frac{\varepsilon_{1,n}}{\sqrt{\delta_n}} = \rho_k(\Xi_n^{a,b}),\ \ N_{x,1} = \mathcal{N}_x(\Xi_n^{a,b}),\ \ N_{y,1} = \mathcal{N}_y(\Xi_n^{a,b})$$

almost surely; the only remaining issue is convergence of the noncompact functionals.

Lemma A.3 (Conditional binomial-to-Poisson convergence)**.** *For every fixed* (a,b),

$$\Xi_n^{a,b} \Rightarrow \Xi^{a,b} \tag{A.13}$$

in the vague topology on Radon point measures on $\mathbb{R}^{2p}$. *Moreover,*

$$(A_n, B_n, \Xi_n^{A_n,B_n}) \Rightarrow (A, B, \Xi^{A,B}). \tag{A.14}$$

Proof. Fix $f \in C_c^+(\mathbb{R}^{2p})$ and set

$$I_n(a,b) = \int_{\mathbb{R}^{2p}} (1-e^{-f(s,t)})q_n^{a,b}(s,t)\ \mathrm{d}s\ \mathrm{d}t.$$

Conditional independence gives

$$\mathbb{E}_{a,b}\exp\{-\Xi_n^{a,b}f\} = \{1 - I_n(a,b)\}^{n-1}.$$

By equation (A.11),

$$(n-1)I_n(a,b) \to \int_{\mathbb{R}^{2p}} (1-e^{-f(s,t)})\lambda_{a,b}(s,t)\ \mathrm{d}s\ \mathrm{d}t,$$

and $I_n(a,b) = O(n^{-1})$. Hence

$$\mathbb{E}_{a,b} e^{-\Xi_n^{a,b} f} \to \exp\left\{ -\int_{\mathbb{R}^{2p}} (1 - e^{-f}) \lambda_{a,b} \right\},$$

the Laplace functional of $\Xi^{a,b}$. Laplace functionals on C_c^+ determine the law and characterize vague convergence of Radon point processes; see, for example, Kallenberg [6], Resnick [12]. This proves equation (A.13).

For convergence jointly with the query values, take bounded continuous $g : \mathbb{R}^{2p} \to \mathbb{R}$. Conditional on (A_n, B_n), the preceding Laplace functional converges at every fixed query and is bounded by one. Dominated convergence against the fixed $N(0, I_p) \otimes N(0, I_p)$ query distribution gives

$$\mathbb{E}\left[g(A_n, B_n) e^{-\Xi_n^{A_n, B_n} f} \right] \to \mathbb{E}\left[g(A, B) e^{-\Xi^{A,B} f} \right].$$

This mixed Laplace-functional class determines the joint law and proves equation (A.14). □

A.3 Gaussian Tail Bounds for Marginal Counts

Vague convergence in equation (A.13) controls bounded regions, but the marginal-count regions are unbounded in the other coordinate. For $0 < R < L$, define the two remote portions of the marginal slabs:

$$E_x(R, L) = \{(s,t) : \|s\|_\infty \le R,\ \|t\|_\infty > L\}, \tag{A.15}$$

$$E_y(R, L) = \{(s,t) : \|t\|_\infty \le R,\ \|s\|_\infty > L\}. \tag{A.16}$$

Let $W \sim N(0, I_p)$, and write

$$\overline{G}_p(q) = \mathbb{P}(\|W\|_\infty > q),$$

with $\overline{G}_p(q) = 1$ for $q < 0$.

Lemma A.4 (Two-sided remote-strip bounds)**.** *For every fixed pair of query values (a, b) and $R < \infty$,*

$$\lim_{L\to\infty} \limsup_{n\to\infty} \mathbb{E}_{a,b} \Xi_n^{a,b}\{E_x(R, L)\} = 0, \tag{A.17}$$

$$\lim_{L\to\infty} \limsup_{n\to\infty} \mathbb{E}_{a,b} \Xi_n^{a,b}\{E_y(R, L)\} = 0. \tag{A.18}$$

The same two assertions hold with $\Xi_n^{a,b}$ replaced by $\Xi^{a,b}$.

Proof. Choose n_0 so that $\delta_n \le 1/2$ for $n \ge n_0$, and set

$$K = \sup_{n \ge n_0} (n-1)\delta_n^{p/2} < \infty,\ C_p = (2\pi)^{-p/2}.$$

Before the deterministic mapping

$$t = (1 - \delta_n)s + \sqrt{1 - \delta_n}(w - b),$$

the mean element of the marked binomial process is

$$(n-1)\delta_n^{p/2} \varphi_p(a + \sqrt{\delta_n} s)\varphi_p(w)\, \mathrm{d}s\, \mathrm{d}w. \tag{A.19}$$

If $\|s\|_\infty \le R$ and $\|t\|_\infty > L$, then

$$\|w\|_\infty > L - R - \|b\|_\infty.$$

Using $\varphi_p \le C_p$ in equation (A.19) therefore gives

$$\mathbb{E}_{a,b}\Xi_n^{a,b}\{E_x(R,L)\} \le KC_p(2R)^p\overline{G}_p(L - R - \|b\|_\infty). \tag{A.20}$$

This tends to zero as $L \to \infty$, uniformly over $n \ge n_0$.

For the reverse strip, fix w. The set of s for which

$$\|(1-\delta_n)s + \sqrt{1-\delta_n}(w-b)\|_\infty \le R$$

is a cube of volume

$$(2R)^p(1-\delta_n)^{-p} \le (4R)^p.$$

If that cube contains an s with $\|s\|_\infty > L$, then

$$\|w\|_\infty > \frac{L}{\sqrt{2}} - \sqrt{2}R - \|b\|_\infty.$$

Integrating first in s in equation (A.19) yields

$$\mathbb{E}_{a,b}\Xi_n^{a,b}\{E_y(R,L)\} \le KC_p(4R)^p\overline{G}_p\left(\frac{L}{\sqrt{2}} - \sqrt{2}R - \|b\|_\infty\right), \tag{A.21}$$

which also tends to zero.

The limiting intensity has the marked representation

$$s \text{ has intensity } u\varphi_p(a)\,\mathrm{d}s,\ t = s + W - b. \tag{A.22}$$

Repeating the same calculation, with KC_p replaced by $u\varphi_p(a)$, proves the two limiting assertions. □

By Markov's inequality, equations (A.17) and (A.18) control the probability of even one remote point, which is exactly what is needed to truncate the strict counts.

A.4 The Selected Radius and Strict Marginal Counts

Lemma A.5 (Critical radius tightness)**.** *For every fixed pair of query values (a,b), the conditional sequence $\rho_k(\Xi_n^{a,b})$ is tight, and $\rho_k(\Xi^{a,b}) < \infty$ almost surely.*

Proof. The limiting mean in the joint cube C_R is

$$L_{a,b}(R) = u\varphi_p(a)\int_{[-R,R]^p} \mathbb{P}\{s + W - b \in [-R,R]^p\}\,\mathrm{d}s. \tag{A.23}$$

It is continuous and strictly increasing: the intensity in equation (A.7) is positive everywhere, so every nonempty joint annulus has positive mean. It also diverges. Indeed, conditional on W, the integral is the volume of the intersection of two radius-R cubes separated by $W - b$. Its ratio to $(2R)^p$ tends to one and is bounded by one. Dominated convergence gives

$$L_{a,b}(R) \sim u\varphi_p(a)(2R)^p. \tag{A.24}$$

For each fixed R, lemma A.3 and the zero intensity of ∂C_R imply

$$\Xi_n^{a,b}(C_R) \Rightarrow \mathrm{Pois}\{L_{a,b}(R)\}.$$

Consequently

$$\lim_{n\to\infty} \mathbb{P}_{a,b}\{\rho_k(\Xi_n^{a,b}) > R\} = \mathbb{P}\{\mathrm{Pois}(L_{a,b}(R)) < k\}.$$

The right side tends to zero as $R \to \infty$, proving tightness and almost-sure finiteness of the limiting radius. □

Lemma A.6 (Genericity of the limiting configuration)**.** *Conditional on every fixed (a, b), the following properties hold almost surely on every bounded cube:*

(i) $\Xi^{a,b}$ *is simple;*

(ii) *distinct points have distinct joint radii* $\max\{\|s\|_\infty, \|t\|_\infty\}$*;*

(iii) *every point has unequal block radii* $\|s\|_\infty \neq \|t\|_\infty$*; and*

(iv) *no point other than the kth joint point has either block radius equal to the kth joint radius.*

Proof. The intensity in equation (A.7) is absolutely continuous and locally finite, so the Poisson process is simple. Each equality excluded in items (ii)–(iv) defines a Lebesgue-null subset of a one- or two-point product space. The first or second factorial moment formula shows that the expected number of violating points or pairs in any bounded cube is zero. Take a countable union over integer cubes. □

Lemma A.7 (Convergence of the noncompact strict counts)**.** *For every fixed pair of query values (a, b),*

$$\left(\rho_k(\Xi_n^{a,b}), \mathcal{N}_x(\Xi_n^{a,b}), \mathcal{N}_y(\Xi_n^{a,b})\right) \Rightarrow \left(\rho_k(\Xi^{a,b}), \mathcal{N}_x(\Xi^{a,b}), \mathcal{N}_y(\Xi^{a,b})\right). \tag{A.25}$$

Proof. Fix $\eta > 0$. By lemma A.5, choose R so that the finite and limiting probabilities of a kth radius exceeding R have limsup at most η. By lemma A.4, choose $L > R$ so that the finite and limiting probabilities of a point in

$$E_x(R, L) \cup E_y(R, L)$$

have limsup at most η.

On the complementary good event, the radius and both full marginal counts are determined by the point measure restricted to the compact cube

$$K_L = [-L, L]^p \times [-L, L]^p.$$

Define the corresponding truncated functional, assigning a fixed dummy value when fewer than k joint points occur by radius R. At every configuration satisfying lemma A.6 and having no atom on the deterministic boundaries of K_L or C_R, this truncated functional is continuous in the vague topology. To verify this directly, note that there are finitely many atoms in K_L. Place disjoint small balls around them. Vague convergence forces the same number of atoms into each ball, so the atoms may be labeled with convergent locations. At a generic configuration, positive gaps determine the radius order, which marginal attains the boundary point's joint radius, and all other marginal-count comparisons. The boundary point's distance in that marginal equals the selected radius. The same point and marginal remain selected under a sufficiently small perturbation, so its

exclusion from that strict marginal count persists. All other comparisons persist by their positive gaps.

The two deterministic boundaries have zero limiting intensity. The continuous-mapping theorem applied to equation (A.13) gives convergence of the truncated vector. The probability that the truncated and full vectors differ is at most $2\eta + o(1)$ for the finite process and 2η for the limiting process. Letting $\eta \downarrow 0$ proves equation (A.25). □

Proof of theorem A.1. The exact metric relation equation (A.12) identifies the finite-sample scaled radius and counts with the three functionals of $\Xi_n^{a,b}$. The conditional assertion follows from lemma A.7. For a bounded continuous test function of the query values and the radius/count vector, condition on the query, apply the conditional convergence at each (a, b), and then use bounded convergence against the fixed Gaussian query distribution. This proves equation (A.8). Finiteness of the limiting radius comes from lemma A.5; finiteness of both limiting marginal counts follows from lemma A.4 or, equivalently, from the exact law proved next. □

A.5 The exact stopped-Poisson count law

Once the first k joint neighbors have been revealed, the points outside the selected cube remain conditionally Poisson, with the original intensity restricted to that region. The following lemma establishes this property at the randomly selected radius.

Lemma A.8 (Post-arrival Poisson property)**.** *Let Π be a Poisson process with deterministic intensity μ on a Polish space E, and let $g : E \to (0, \infty)$ be measurable. Suppose $\mu\{g \le r\} < \infty$ for every r, the pushforward $\mu \circ g^{-1}$ is diffuse, and $\mu\{g \le r\} \to \infty$. If τ_k is the kth smallest g-value, then, conditional on the stopped sigma-field generated by points with $g \le \tau_k$, the restriction of Π to $\{g > \tau_k\}$ is Poisson with intensity $\mu|_{\{g>\tau_k\}}$.*

Proof. Let

$$\mathcal{F}_r = \sigma\{\Pi|_{\{g\le s\}} : s \le r\}.$$

Then τ_k is an $(\mathcal{F}_r)$-stopping time. We use the stopped sigma-field

$$\mathcal{F}_{\tau_k} = \{H \in \mathcal{F}_\infty : H \cap \{\tau_k \le r\} \in \mathcal{F}_r \text{ for every } r \ge 0\},\ \mathcal{F}_\infty = \sigma\left(\bigcup_{r\ge 0} \mathcal{F}_r\right).$$

The conditional assertion is a statement about a random intensity restriction. More explicitly, for every nonnegative measurable f whose support has finite intensity,

$$\mathbb{E}\left[e^{-\int_{\{g>\tau_k\}} f \, d\Pi} \,\middle|\, \mathcal{F}_{\tau_k}\right] = \exp\left[-\int_{\{g>\tau_k\}} (1 - e^{-f}) \, d\mu\right].$$

First let σ be a stopping time taking values in a deterministic countable grid. On each event $\{\sigma = r_j\} \in \mathcal{F}_{r_j}$, deterministic-set independent increments imply that $\Pi|_{\{g>r_j\}}$ is a Poisson process with the restricted intensity and is independent of $\mathcal{F}_{r_j}$. Summing over j proves the claimed conditional Laplace-functional identity at σ.

Now round τ_k upward:

$$\tau_k^{(m)} = 2^{-m}\lceil 2^m \tau_k \rceil.$$

These are grid-valued stopping times and decrease to τ_k. If

$$M(r) = \mu\{g \le r\},$$

then M is continuous. On every bounded interval it is uniformly continuous, so for every $R < \infty$,

$$\sup_{0 \le r \le R} \mu\{r < g \le r + 2^{-m}\} \longrightarrow 0.$$

The shell is selected from the process, so this deterministic intensity bound needs one additional step. On $\{\tau_k \le R\}$, a point in $\{\tau_k < g \le \tau_k^{(m)}\}$ produces an ordered pair of radial points in $[0, R+1]$ whose separation is at most 2^{-m}. With $\nu = \mu \circ g^{-1}$, the second factorial-moment formula bounds the probability of such a pair by

$$\nu([0, R+1]) \sup_{0 \le r \le R+1} \nu((r, r + 2^{-m}]) \longrightarrow 0.$$

Thus the adaptively selected intervening shell is empty with probability tending to one. Its intensity also tends to zero uniformly on $\{\tau_k \le R\}$. For every bounded $\mathcal{F}_{\tau_k}$-measurable variable H, H is also $\mathcal{F}_{\tau_k^{(m)}}$-measurable. Multiply the grid-time conditional Laplace identity by H, let $m \to \infty$, and use both the vanishing shell probability and the vanishing change in the restricted intensity's Laplace exponent to pass to τ_k. Then let $R \to \infty$, using $\tau_k < \infty$ almost surely. This proves the post-τ_k conditional Laplace functional for every nonnegative test function with finite intensity support. A monotone-class argument completes the proof. This is also the usual strong Markov property of a radially marked Poisson process; see Last and Penrose [8, Chs. 5 and 7]. □

Proposition A.9 (Exact Gamma–Poisson representation)**.** *Condition on $(A, B) = (a, b)$, and set*

$$c = u\varphi_p(a), \;\; h = c^{1/p}. \tag{A.26}$$

Let $W \sim N(0, I_p)$, and define

$$\Lambda_{h,b}(r) = \prod_{\ell=1}^{p} \mathbb{E}\big[(2r - h|W_\ell - b_\ell|)_+\big]. \tag{A.27}$$

Let $G_k \sim \mathrm{Gamma}(k, 1)$, and let R_k be the unique solution of

$$\Lambda_{h,b}(R_k) = G_k. \tag{A.28}$$

Conditional on (a, b, G_k, R_k), let

$$P_x, P_y \overset{\text{ind}}{\sim} \mathrm{Pois}\{(2R_k)^p - G_k\}, \;\; J \sim \mathrm{Bernoulli}(1/2),$$

with J independent of P_x, P_y, G_k, R_k. Then

$$h\rho_k(\Xi^{a,b}) \overset{d}{=} R_k, \tag{A.29}$$

and

$$\big(\mathcal{N}_x(\Xi^{a,b}), \mathcal{N}_y(\Xi^{a,b})\big) \overset{d}{=} \begin{cases} \big(k - 1 + P_x, \;\; k + P_y\big), & J = 0, \\ \big(k + P_x, \;\; k - 1 + P_y\big), & J = 1, \end{cases} \tag{A.30}$$

In particular,

$$\mathbb{P}\big(\{\mathcal{N}_x, \mathcal{N}_y\} = \{k-1, k\} \mid a, b, G_k, R_k\big) = \exp[-2\{(2R_k)^p - G_k\}]. \tag{A.31}$$

Proof. The intensity in equation (A.7) can be constructed by taking a homogeneous Poisson process of rate c in the s-coordinate, attaching iid $N(0, I_p)$ marks W, and mapping

$$t = s + W - b.$$

Scale both coordinate blocks by $h = c^{1/p}$:

$$S' = hs,\ T' = ht = S' + h(W - b). \tag{A.32}$$

Then S' has unit Lebesgue intensity. The T'-marginal also has unit Lebesgue intensity because

$$\int_{\mathbb{R}^p} h^{-p} \varphi_p\left(b + \frac{t' - s'}{h}\right)\, \mathrm{d}s' = 1.$$

For the normalized joint cube

$$B_r = \{(s', t') : \|s'\|_\infty \le r,\ \|t'\|_\infty \le r\},$$

condition on W. The admissible s'-set is the intersection of two radius-r cubes separated coordinatewise by $h(W - b)$; its volume is

$$\prod_{\ell=1}^{p} (2r - h|W_\ell - b_\ell|)_+.$$

Taking expectations and using independence of the Gaussian coordinates gives equation (A.27). The function is continuous by dominated convergence on bounded r-intervals. It is strictly increasing because the intensity is positive on every nonempty joint annulus. Finally,

$$\frac{\Lambda_{h,b}(r)}{(2r)^p} \to 1$$

by dominated convergence, so it tends to infinity.

Map every normalized point to its joint radius

$$g(s', t') = \max\{\|s'\|_\infty, \|t'\|_\infty\}.$$

The Poisson mapping theorem [8] gives an inhomogeneous Poisson process on $(0, \infty)$ with cumulative mean $\Lambda_{h,b}(r)$. Its intensity is diffuse. Time change by $\Lambda_{h,b}$ produces a unit-rate Poisson process, whose kth arrival is Gamma$(k, 1)$. This proves equations (A.28) and (A.29).

Apply lemma A.8 at R_k. The two marginal-only regions

$$D_x(r) = \{\|S'\|_\infty < r,\ \|T'\|_\infty > r\},$$
$$D_y(r) = \{\|T'\|_\infty < r,\ \|S'\|_\infty > r\}$$

are disjoint and lie strictly outside B_r. Since both normalized marginals have unit intensity,

$$\mu\{D_x(r)\} = \mu\{D_y(r)\} = (2r)^p - \Lambda_{h,b}(r).$$

The conditional exterior-process law also applies to these radius-dependent sets: first restrict to $R_k \le L$ and intersect the sets with a deterministic compact box. Approximation by measurable simple functions of the radius extends the conditional Laplace identity to these sets; the radial boundaries have zero intensity. Monotone convergence as the box expands, followed by $L \uparrow \infty$,

gives the same identity for the full strips, whose displayed intensities are finite. Conditional on the stopped history at R_k, their counts are therefore independent Poisson variables with mean

$$(2R_k)^p - \Lambda_{h,b}(R_k) = (2R_k)^p - G_k.$$

Their conditional law depends on the stopped history only through R_k. Conditional on that radius and the fixed query values, they are therefore independent of the boundary point.

It remains to determine the active block of the unique boundary point. The conditional joint intensity is not invariant under the bare swap $(s', t') \mapsto (t', s')$ when $b \neq 0$. It is invariant under the sign-swap

$$(s', t') \longmapsto (-t', -s'),$$

because $t' - s'$ is unchanged. This map preserves every symmetric joint cube and exchanges source-active with output-active faces. Let ν_x and ν_y be the pushforward intensity measures on the radius coordinate for boundary points whose active block is, respectively, source and output. The sign-swap gives $\nu_x = \nu_y$. The set $\{\|s'\|_\infty = \|t'\|_\infty\}$ has zero intensity, so $\nu_x + \nu_y = \nu$, the full radial intensity measure. Therefore the pushforward intensity on radius and orientation factorizes as $\nu \otimes \text{Bernoulli}(1/2)$. The Poisson marking theorem then makes the orientations iid fair marks, independent of the radial process. Together with the stopped exterior law, this makes the kth orientation independent of (R_k, P_x, P_y), conditional on the query values.

There are $k - 1$ earlier joint points strictly inside both marginal cubes. The boundary point is excluded from the strict count in its active block and included in the other. Adding the two marginal-only Poisson counts proves equation (A.30). Saturation occurs exactly when $P_x = P_y = 0$, whose conditional probability is the right side of equation (A.31). □

Together, theorem A.1 and proposition A.9 prove theorem 5.2 part (b) and the exact functional stated in proposition 3.1.

Remark A.10 (Anchor sign convention). With the actual query noise $B = Z_1$, the exact conditional intensity is

$$u\varphi_p(A)\varphi_p(B + T - S),$$

and the normalized marked relation is $T' = S' + h(W - B)$. If instead $V = -Z_1$, Gaussian evenness rewrites the density as

$$u\varphi_p(A)\varphi_p(V + S - T).$$

Since $V \overset{d}{=} B$, the unconditional mixture is the same, but a conditional theorem must not use the latter density while declaring $V = Z_1$.

A.6 Finite-Sample Score Moments

The next lemma controls the counts in the finite sample, not just their Poisson limit. Bounds on the limiting variables alone would not rule out rare finite-sample neighborhoods with counts of order n.

Lemma A.11 (Uniform fractional count moments). *Under equation (A.1), for every $0 < \alpha < 1$,*

$$\sup_{n>k} \mathbb{E}\{(N_{x,1} + 1)^\alpha + (N_{y,1} + 1)^\alpha\} < \infty. \tag{A.33}$$

More precisely, for every $0 < \beta < 1$, a finite C_β exists such that, for all integers $m \geq 1$,

$$\sup_{n>k} \max\{\mathbb{P}(N_{x,1} \geq m), \mathbb{P}(N_{y,1} \geq m)\} \leq C_\beta m^{-\beta}. \tag{A.34}$$

Proof. We bound a large marginal count by splitting according to whether the selected joint radius exceeds a deterministic threshold. Below that threshold, a binomial tail bound controls the marginal count; above it, we bound the probability that the joint ball contains fewer than k points. It suffices to work for sufficiently large n, because finitely many earlier sample sizes have bounded counts. Choose n_0 so that, for $n \geq n_0$,

$$\delta_n \leq \frac{1}{2},\ 0 < \underline{u} \leq (n-1)\delta_n^{p/2} \leq \overline{u} < \infty. \tag{A.35}$$

Increase $\overline{u}$, if necessary, so that $u_n \leq \overline{u}$ as well. Let $C_p = (2\pi)^{-p/2}$.

Conditional on $(A_n, B_n) = (a, b)$, define

$$Q_{x,n}(r) = \sum_{j=2}^{n} \mathbf{1}\{\|\xi_{j,n}\|_\infty \leq r\},\ Q_{y,n}(r) = \sum_{j=2}^{n} \mathbf{1}\{\|\eta_{j,n}\|_\infty \leq r\}.$$

The ξ-density is $\delta_n^{p/2}\varphi_p(a + \sqrt{\delta_n}s)$. The other output samples are independent of the query and have law $N(0, (1-\delta_n)I_p)$. After centering at $Y_{1,n}$ and scaling by $\sqrt{\delta_n}$, the conditional η-density is bounded by

$$\delta_n^{p/2}(1-\delta_n)^{-p/2}C_p.$$

Thus a deterministic $K < \infty$ exists such that, for every query, $r > 0$, and $n \geq n_0$,

$$\max\{\mathbb{E}_{a,b}Q_{x,n}(r), \mathbb{E}_{a,b}Q_{y,n}(r)\} \leq Kr^p. \tag{A.36}$$

Choose $\gamma > 0$ so small that $K\gamma^p \leq 1/4$, and for $1 \leq m \leq n-1$ put

$$r_m = \gamma m^{1/p}.$$

Both Q-variables are binomial conditional on the query, and equation (A.36) bounds their means at r_m by $m/4$. The elementary binomial Chernoff bound therefore gives

$$\max\{\mathbb{P}_{a,b}(Q_{x,n}(r_m) \geq m), \mathbb{P}_{a,b}(Q_{y,n}(r_m) \geq m)\} \leq (e/4)^m. \tag{A.37}$$

Write $\rho_{n,k} = \varepsilon_{1,n}/\sqrt{\delta_n}$. If $\rho_{n,k} \leq r_m$, every point counted by $N_{x,1}$ is counted by $Q_{x,n}(r_m)$, and similarly for the output block. Hence

$$\max\{\mathbb{P}_{a,b}(N_{x,1} \geq m), \mathbb{P}_{a,b}(N_{y,1} \geq m)\} \leq (e/4)^m + \mathbb{P}_{a,b}(\rho_{n,k} > r_m). \tag{A.38}$$

To bound the joint radius, let $W \sim N(0, I_p)$ and set

$$q_p = \mathbb{P}(\|W\|_\infty \leq 1) > 0.$$

Consider points satisfying

$$\|\xi_{j,n}\|_\infty \leq r_m/4,\ \|Z_j\|_\infty \leq 1. \tag{A.39}$$

If $r_m \geq 2(1 + \|b\|_\infty)$, then equation (A.9) gives

$$\|\eta_{j,n}\|_\infty \leq r_m/4 + (1 + \|b\|_\infty) \leq 3r_m/4 < r_m.$$

Every favorable point is therefore inside the joint radius-r_m cube.

Because $m \leq n-1$,

$$\sqrt{\delta_n}r_m \leq \gamma(n\delta_n^{p/2})^{1/p} \leq \gamma\overline{u}^{1/p}.$$

Put

$$H = \frac{\sqrt{p}\,\gamma \overline{u}^{1/p}}{4}.$$

On the source cube in equation (A.39),

$$\varphi_p(a + \sqrt{\delta_n} s) \geq C_p \exp[-(\|a\|_2 + H)^2/2].$$

The number of favorable points is consequently binomial with mean at least

$$\lambda_m(a) \geq d_0 m \exp[-(\|a\|_2 + H)^2/2],\ d_0 = \frac{\underline{u}\, q_p C_p \gamma^p}{2^p} > 0. \tag{A.40}$$

If B_0 is any binomial variable of mean λ, then for fixed k there is a finite D_k such that

$$\mathbb{P}(B_0 < k) \leq D_k e^{-\lambda/2}. \tag{A.41}$$

Indeed, if $B_0 \sim \mathrm{Bin}(N, q)$, then for $0 \leq j < k$,

$$\binom{N}{j} q^j (1-q)^{N-j} \leq e^k e^{-Nq} (Nq)^j / j!,$$

and $\sum_{j<k} \lambda^j / j!$ is bounded by a constant times $e^{\lambda/2}$.

The event $\rho_{n,k} > r_m$ means that the full joint cube contains fewer than k points, and therefore implies that the favorable count is below k. From equations (A.40) and (A.41),

$$\begin{aligned} \mathbb{P}_{a,b}(\rho_{n,k} > r_m) \leq\ & \mathbf{1}\{r_m < 2(1 + \|b\|_\infty)\} \\ & + D_k \exp\left[-d_1 m e^{-(\|a\|_2 + H)^2/2}\right], \end{aligned} \tag{A.42}$$

where $d_1 = d_0/2$.

Average over the independent standard Gaussian query values A, B. Gaussian tails imply, after enlarging constants to cover small m,

$$\mathbb{P}\{\gamma m^{1/p} < 2(1 + \|B\|_\infty)\} \leq C e^{-c m^{2/p}}. \tag{A.43}$$

For the remaining mixture, fix $0 < \beta < 1$, select $\beta < \beta_1 < 1$, and let

$$R_m = (2\beta_1 \log m)^{1/2},\ m \geq 2.$$

On $\{\|A\|_2 \leq R_m\}$,

$$m e^{-(\|A\|_2 + H)^2/2} \geq m^{1-\beta_1} e^{-H(2\beta_1 \log m)^{1/2} - H^2/2} \geq m^{(1-\beta_1)/2}$$

for all sufficiently large m. On the complement, choose $\theta \in (\beta/(2\beta_1), 1/2)$. Since $\mathbb{E} e^{\theta \|A\|_2^2} = (1-2\theta)^{-p/2}$,

$$\mathbb{P}(\|A\|_2 > R_m) \leq (1-2\theta)^{-p/2} e^{-\theta R_m^2} \leq C_\beta m^{-\beta}.$$

Splitting at R_m therefore gives

$$\mathbb{E} \exp\left[-d_1 m e^{-(\|A\|_2 + H)^2/2}\right] \leq C_\beta m^{-\beta}. \tag{A.44}$$

Combining equations (A.38) and (A.42) to (A.44) proves equation (A.34) for all sufficiently large n. The finitely many earlier n's and the case $m > n - 1$ are absorbed into C_β.

Finally, choose $0 < \alpha < \beta < 1$. For either marginal count N,

$$\begin{aligned}\mathbb{E}(N+1)^{\alpha} &= 1 + \sum_{m\geq 1}\{(m+1)^{\alpha} - m^{\alpha}\}\mathbb{P}(N \geq m) \\ &\leq 1 + C\sum_{m\geq 1} m^{\alpha-1-\beta} < \infty\end{aligned}$$

uniformly in n. This proves equation (A.33). □

Corollary A.12 (Uniform integrability of the neighborhood score)**.** *Let*

$$H_n = \psi(N_{x,1}+1) + \psi(N_{y,1}+1)$$

and

$$H = \psi\{\mathcal{N}_x(\Xi^{A,B})+1\} + \psi\{\mathcal{N}_y(\Xi^{A,B})+1\}.$$

By proposition A.9, H has the same distribution as $\psi(M_x+1)+\psi(M_y+1)$. Then $\{H_n^2\}$ is uniformly integrable and

$$\mathbb{E}H_n \to \mathbb{E}H,\ \mathbb{E}H_n^2 \to \mathbb{E}H^2. \tag{A.45}$$

The same assertions hold when each digamma term is replaced by the logarithm of its integer argument.

Proof. Fix $\alpha \in (0,1)$ and $q > 1$. Every fixed power of $\log(m+1)$ is bounded by a constant times $(m+1)^{\alpha}$. Hence lemma A.11 and $(x+y)^{2q} \leq 2^{2q-1}(x^{2q}+y^{2q})$ imply

$$\sup_n \mathbb{E}\left|\log(N_{x,1}+1) + \log(N_{y,1}+1)\right|^{2q} < \infty.$$

For integers $m \geq 0$,

$$|\psi(m+1) - \log(m+1)| \leq C$$

with a universal finite constant C. Thus $\sup_n \mathbb{E}|H_n|^{2q} < \infty$, which implies uniform integrability of $\{H_n^2\}$. Theorem A.1 gives $H_n \Rightarrow H$. Uniform integrability yields convergence of the first and second moments in equation (A.45). The logarithmic version was already proved by the same L^{2q} bound. □

B Thin and regular finite-sample endpoints

We prove the thin-count collapse and regular radius/count scaling used in theorem 5.2 directly for the finite Gaussian array.

B.1 Conditional local coordinates

It suffices initially to study query observation 1. Put $m_n = n-1$, condition on

$$U = X_1 = a,\ V = Z_1 = b,$$

and let φ_p be the standard p-variate Gaussian density. Define

$$\alpha_n = \{m_n\varphi_p(a)\}^{1/p},\ h_n = \alpha_n\sqrt{\delta_n(1-\delta_n)}. \tag{B.1}$$

For $j = 2, \dots, n$, introduce

$$\mathsf{S}_{j,n} = \alpha_n(X_j - a),\ \mathsf{T}_{j,n} = \alpha_n(Y_{j,n} - Y_{1,n}). \tag{B.2}$$

The channel identity equation (1) gives, exactly,

$$\mathsf{T}_{j,n} = (1 - \delta_n)\mathsf{S}_{j,n} + h_n(Z_j - b). \tag{B.3}$$

Write

$$\mathsf{A}_{j,n} = \|\mathsf{S}_{j,n}\|_\infty,\ \mathsf{B}_{j,n} = \|\mathsf{T}_{j,n}\|_\infty,\ \mathsf{J}_{j,n} = \mathsf{A}_{j,n} \vee \mathsf{B}_{j,n},$$

and let $\rho_n = \alpha_n \varepsilon_{1,n}$, the kth order statistic of $\{\mathsf{J}_{j,n} : 2 \le j \le n\}$. The common positive rescaling does not change the strict marginal counts:

$$N_{x,1} = \sum_{j=2}^{n} \mathbf{1}\{\mathsf{A}_{j,n} < \rho_n\},\ N_{y,1} = \sum_{j=2}^{n} \mathbf{1}\{\mathsf{B}_{j,n} < \rho_n\}. \tag{B.4}$$

For $r \ge 0$, define the conditional mean measures

$$\begin{aligned} M_{x,n}(r; a, b) &= m_n \mathbb{P}(\mathsf{A}_{2,n} \le r \mid U = a, V = b), \\ M_{y,n}(r; a, b) &= m_n \mathbb{P}(\mathsf{B}_{2,n} \le r \mid U = a, V = b), \\ \Lambda_n(r; a, b) &= m_n \mathbb{P}(\mathsf{J}_{2,n} \le r \mid U = a, V = b). \end{aligned} \tag{B.5}$$

The conditioning on b is redundant in the first line but is retained to keep the notation uniform.

Lemma B.1 (Gaussian general position)**.** *For every $n > k$ and $0 < \delta_n < 1$, conditionally on every finite query pair (a, b), the variables $\{\mathsf{J}_{j,n} : 2 \le j \le n\}$ are almost surely distinct, and*

$$\mathbb{P}(\mathsf{A}_{j,n} = \mathsf{B}_{j,n} \mid U = a, V = b) = 0.$$

Consequently, the kth joint shell has one boundary observation and that observation is strictly inside exactly one of the two projected marginal balls.

Proof. For $0 < \delta_n < 1$, the linear map

$$(X_j, Z_j) \longmapsto (\mathsf{S}_{j,n}, \mathsf{T}_{j,n})$$

has a nonsingular block-triangular matrix: its first diagonal block is $\alpha_n I_p$, and its second is $h_n I_p$. Hence $(\mathsf{S}_{j,n}, \mathsf{T}_{j,n})$ has a Lebesgue density on $\mathbb{R}^{2p}$, conditionally on (a, b).

After choosing the coordinates and signs attaining the two max norms, the event $\|\mathsf{S}_{j,n}\|_\infty = \|\mathsf{T}_{j,n}\|_\infty$ lies in a finite union of nontrivial affine hyperplanes and therefore has Lebesgue measure zero. The same coordinate-and-sign decomposition shows that equality between $\mathsf{J}_{j,n}$ and $\mathsf{J}_{\ell,n}$, for $j \ne \ell$, lies in a finite union of null hypersurfaces in the product sample. A finite union over pairs j, ℓ remains null. The last assertion follows because the unique boundary observation satisfies $\mathsf{A}_{j,n} \vee \mathsf{B}_{j,n} = \rho_n$, while the two block radii are unequal. □

Lemma B.2 (Monotone local-uniform convergence)**.** *Let $f_n : [0, \infty) \to \mathbb{R}$ be nondecreasing. If $f_n(r) \to f(r)$ at every $r \ge 0$, where f is continuous, then, for every $M < \infty$,*

$$\sup_{0 \le r \le M} |f_n(r) - f(r)| \longrightarrow 0.$$

Proof. Fix $\epsilon > 0$. Uniform continuity of f on $[0, M]$ gives a partition

$$0 = r_0 < r_1 < \cdots < r_L = M$$

such that $f(r_\ell) - f(r_{\ell-1}) < \epsilon$ for every ℓ. Pointwise convergence at the finitely many grid points gives an n_0 such that

$$\max_{0 \le \ell \le L} |f_n(r_\ell) - f(r_\ell)| < \epsilon$$

for $n \ge n_0$. If $r \in [r_{\ell-1}, r_\ell]$, monotonicity gives

$$f_n(r_{\ell-1}) \le f_n(r) \le f_n(r_\ell).$$

The upper and lower bounds differ from $f(r)$ by at most 2ϵ. Thus the required supremum is at most 2ϵ for $n \ge n_0$. Letting $\epsilon \downarrow 0$ proves the result. □

Lemma B.3 (Conditional decomposition at the kth radius)**.** *Let (A_j, B_j), $1 \le j \le m$, be iid with a joint density. Put $D_j = A_j \vee B_j$, suppose D_j has a continuous density, and let $R = D_{(k)}$. Define*

$$q_x(r) = \mathbb{P}(A < r, B > r), \; q_y(r) = \mathbb{P}(B < r, A > r).$$

Conditionally on $R = r$, the boundary observation, and the labels of the observations above and below the boundary, the $m - k$ upper observations are independent with the original law conditioned on $D > r$. After averaging over those labels, their expected total number entering either strict marginal ball is

$$(m - k)\frac{q_x(r) + q_y(r)}{1 - F_D(r)}. \tag{B.6}$$

If additionally $\mathbb{P}(A = B) = 0$, the unordered strict-count pair at radius R is $\{k - 1, k\}$ if and only if no above-boundary observation enters either marginal ball.

Proof. Let $f_{A,B}$ be the joint density and f_D the density of D. The density of $R = D_{(k)}$ is

$$f_R(r) = \frac{m!}{(k-1)!(m-k)!} F_D(r)^{k-1}\{1 - F_D(r)\}^{m-k} f_D(r). \tag{B.7}$$

To see the conditional factorization explicitly, first choose which $k - 1$ observations lie below the boundary and which observation lies on the boundary. Before integrating the individual observations, the corresponding joint density contains $k - 1$ factors proportional to

$$f_{A,B}(s, t)\mathbf{1}\{s \vee t < r\}$$

and $m - k$ factors proportional to

$$f_{A,B}(s, t)\mathbf{1}\{s \vee t > r\}.$$

For any fixed choice of labels, conditioning on these labels and the boundary radius normalizes each lower factor by $F_D(r)$ and each upper factor by $1 - F_D(r)$; the combinatorial prefactor in equation (B.7) accounts for the possible label choices. This constructs a regular conditional distribution under which the labelled upper observations are conditionally iid with the law of (A, B) given $D > r$.

An upper observation enters the x-marginal ball precisely on $\{A < r, B > r\}$, and enters the y-marginal ball precisely on $\{B < r, A > r\}$, up to null boundary events. Summing these two conditional probabilities over the $m - k$ upper observations proves equation (B.6).

Exactly $k-1$ observations have $D < r$, and each lies strictly inside both marginal balls. The unique boundary observation has $A \vee B = r$. When $A \neq B$, it lies on one marginal boundary and strictly inside the other, so it contributes to exactly one strict count. Hence the base unordered pair is $\{k-1, k\}$, and it remains that pair exactly when no upper observation contributes an additional marginal count. □

B.2 The Thin Regime

Lemma B.4 (Diagonal conditional small-ball limit)**.** *If $u_n = n\delta_n^{p/2} \to 0$, then, for every finite pair of query values (a, b) and every $M < \infty$,*

$$\sup_{0 \le r \le M} |M_{x,n}(r; a, b) - (2r)^p| \to 0, \tag{B.8}$$

$$\sup_{0 \le r \le M} |M_{y,n}(r; a, b) - (2r)^p| \to 0, \tag{B.9}$$

$$\sup_{0 \le r \le M} |\Lambda_n(r; a, b) - (2r)^p| \to 0. \tag{B.10}$$

In particular,

$$\sup_{0 \le r \le M} \{M_{x,n}(r) - \Lambda_n(r)\} + \sup_{0 \le r \le M} \{M_{y,n}(r) - \Lambda_n(r)\} \longrightarrow 0. \tag{B.11}$$

Proof. The assumption $u_n \to 0$ implies $\delta_n \to 0$. For fixed a,

$$\alpha_n \longrightarrow \infty$$

and, by equation (B.1),

$$h_n^p = m_n \varphi_p(a) \delta_n^{p/2} (1-\delta_n)^{p/2} = \frac{m_n}{n} \varphi_p(a) u_n (1-\delta_n)^{p/2} \longrightarrow 0. \tag{B.12}$$

Let $C_r = [-r, r]^p$. The change of variables $s = \alpha_n(x - a)$ gives

$$M_{x,n}(r; a, b) = \frac{1}{\varphi_p(a)} \int_{C_r} \varphi_p(a + s/\alpha_n)\, \mathrm{d}s \longrightarrow (2r)^p. \tag{B.13}$$

The output marginal is $N(0, (1-\delta_n) I_p)$. Denote its density by

$$g_{\delta_n}(y) = (1-\delta_n)^{-p/2} \varphi_p\left(\frac{y}{\sqrt{1-\delta_n}} \right)$$

and the fixed query output by

$$y_{1,n}(a, b) = (1-\delta_n) a + \sqrt{\delta_n (1-\delta_n)}\, b.$$

Then $y_{1,n}(a, b) \to a$, and the change of variables $t = \alpha_n(y - y_{1,n})$ gives

$$M_{y,n}(r; a, b) = \frac{1}{\varphi_p(a)} \int_{C_r} g_{\delta_n}(y_{1,n} + t/\alpha_n)\, \mathrm{d}t \longrightarrow (2r)^p. \tag{B.14}$$

For the joint ball, let $W \sim N(0, I_p)$. From equation (B.3),

$$\Lambda_n(r; a, b) = \frac{1}{\varphi_p(a)} \int_{C_r} \varphi_p(a + s/\alpha_n) \mathbb{P}\big((1-\delta_n)s + h_n(W - b) \in C_r\big)\, \mathrm{d}s. \tag{B.15}$$

For every s in the interior of C_r, the random vector inside the probability converges in probability to s, by equation (B.12); hence the probability tends to one. The boundary of C_r has Lebesgue measure zero. On a fixed box, $\varphi_p(a + s/\alpha_n)/\varphi_p(a)$ is bounded for all sufficiently large n, while the probability is bounded by one. Dominated convergence therefore proves

$$\Lambda_n(r; a, b) \longrightarrow (2r)^p$$

for every fixed r. The joint-ball integral includes all Gaussian noise values, and the output-marginal calculation includes all source locations; neither calculation truncates the noise distribution.

Each mean measure is nondecreasing in r, and their common pointwise limit is continuous. Lemma B.2 proves equations (B.8) to (B.10). A joint ball is contained in each marginal ball, so both differences in equation (B.11) are nonnegative. The three uniform convergences then imply equation (B.11). □

Theorem B.5 (Saturation in the thin regime)**.** *If $u_n \to 0$, then*

$$\mathbb{P}\{\{N_{x,1}, N_{y,1}\} = \{k-1, k\}\} \longrightarrow 1. \tag{B.16}$$

Consequently,

$$\overline{S}_{n,k} \longrightarrow 1 \text{ in } L^1 \tag{B.17}$$

and

$$\frac{\widehat{I}_{n,k}}{\log n} \longrightarrow 1 \text{ in } L^1. \tag{B.18}$$

Proof. Fix finite query values (a, b). Let $F_n(\,\cdot\,; a, b)$ be the conditional CDF of one $\mathsf{J}_{j,n}$. By definition and lemma B.4,

$$m_n F_n(r; a, b) = \Lambda_n(r; a, b) \longrightarrow (2r)^p.$$

The binomial order-statistic identity gives, for $r > 0$,

$$\begin{aligned}\mathbb{P}(\rho_n \le r \mid a, b) &= \mathbb{P}\{\mathrm{Bin}(m_n, F_n(r; a, b)) \ge k\} \\ &\longrightarrow \mathbb{P}\{\mathrm{Pois}((2r)^p) \ge k\} = \mathbb{P}(G_k \le (2r)^p),\end{aligned} \tag{B.19}$$

where $G_k \sim \mathrm{Gamma}(k, 1)$. Therefore

$$\rho_n \Rightarrow R_0 := \frac{1}{2} G_k^{1/p} \text{ conditionally on } (a, b). \tag{B.20}$$

In particular, ρ_n is conditionally tight.

For deterministic r, define

$$\begin{aligned}q_{x,n}(r) &= \mathbb{P}(\mathsf{A}_{2,n} < r, \mathsf{B}_{2,n} > r \mid a, b), \\ q_{y,n}(r) &= \mathbb{P}(\mathsf{B}_{2,n} < r, \mathsf{A}_{2,n} > r \mid a, b).\end{aligned}$$

All relevant distributions are continuous, so strict and non-strict versions agree, and

$$m_n q_{x,n}(r) = M_{x,n}(r) - \Lambda_n(r), \;\; m_n q_{y,n}(r) = M_{y,n}(r) - \Lambda_n(r). \tag{B.21}$$

Fix $M < \infty$, and let $\mathcal{E}_n$ be the event that an observation above the kth joint boundary enters one of the two strict marginal balls. The local-uniform convergence in lemma B.4 gives

$$\sup_{r \le M} m_n\{q_{x,n}(r) + q_{y,n}(r)\} \longrightarrow 0.$$

It also gives

$$\sup_{r\le M} F_n(r) = \frac{\Lambda_n(M)}{m_n} = O(m_n^{-1}),$$

so $1-F_n(r) \ge 1/2$ uniformly on $[0, M]$ for all sufficiently large n. Applying lemma B.3 conditionally at ρ_n, followed by Markov's inequality, yields

$$\mathbb{P}(\mathcal{E}_n, \rho_n \le M \mid a, b) \le 2 \sup_{0\le r\le M} m_n\{q_{x,n}(r) + q_{y,n}(r)\} \longrightarrow 0. \tag{B.22}$$

On $\mathcal{E}_n^c$, the $k-1$ observations below the joint boundary enter both marginal balls, and the unique boundary observation enters exactly one by lemma B.1. The count pair is then $\{k-1, k\}$.

Given $\epsilon > 0$, conditional tightness in equation (B.20) permits a finite M for which

$$\limsup_n \mathbb{P}(\rho_n > M \mid a, b) < \epsilon.$$

With this M, equation (B.22) shows that the conditional probability of a nonsaturated neighborhood is at most $\epsilon + o(1)$. Let $\epsilon \downarrow 0$. This proves equation (B.16) conditionally for every finite (a, b). Since (U, V) is finite almost surely and conditional failure probabilities lie in $[0, 1]$, dominated convergence removes the conditioning and proves equation (B.16).

Exchangeability and $0 \le \overline{S}_{n,k} \le 1$ give

$$\mathbb{E}|1 - \overline{S}_{n,k}| = \mathbb{E}(1 - \overline{S}_{n,k}) = \mathbb{P}(S_{1,n} = 0) \longrightarrow 0,$$

which is equation (B.17).

It remains to prove the estimator conclusion, which does not follow merely from convergence for one neighborhood unless the possible contribution of a nonsaturated neighborhood is bounded. General position implies, for every query,

$$N_{x,i} \ge k-1,\ N_{y,i} \ge k-1,\ \max(N_{x,i}, N_{y,i}) \ge k.$$

Because ψ is increasing, the neighborhood's deficit

$$\mathcal{L}_{i,n} = \psi(N_{x,i}+1) + \psi(N_{y,i}+1) - \psi(k) - \psi(k+1) \tag{B.23}$$

is nonnegative and vanishes exactly on a saturated neighborhood. Since both counts are at most $n-1$,

$$0 \le \mathcal{L}_{i,n} \le H_{n,k} := 2\psi(n) - \psi(k) - \psi(k+1). \tag{B.24}$$

Substitution into the exact statistic equation (4) gives

$$C_{n,k} - \widehat{I}_{n,k} = \frac{1}{n}\sum_{i=1}^{n} \mathcal{L}_{i,n},\ C_{n,k} = \psi(n) - \psi(k+1). \tag{B.25}$$

By exchangeability, equation (B.16), and equation (B.24),

$$\begin{aligned} 0 \le \mathbb{E}\frac{C_{n,k} - \widehat{I}_{n,k}}{\log n} &= \frac{\mathbb{E}\mathcal{L}_{1,n}}{\log n} \\ &\le \frac{H_{n,k}}{\log n}\mathbb{P}(S_{1,n} = 0) \longrightarrow 0, \end{aligned} \tag{B.26}$$

because $H_{n,k}/\log n \to 2$ for fixed k. Also $C_{n,k}/\log n \to 1$. Since the difference in equation (B.25) is nonnegative,

$$\left|\frac{\widehat{I}_{n,k}}{\log n} - 1\right| \le \left|\frac{C_{n,k}}{\log n} - 1\right| + \frac{C_{n,k} - \widehat{I}_{n,k}}{\log n}.$$

Taking expectations and using equation (B.26) proves equation (B.18). □

B.3 The Regular Regime

The regular result concerns a shrinking channel. We therefore assume both

$$\delta_n \longrightarrow 0,\ u_n = n\delta_n^{p/2} \longrightarrow \infty. \tag{B.27}$$

The first condition is not implied by the second. Put

$$b_n = u_n^{1/(2p)}.$$

Lemma B.6 (Regular conditional mean measures)**.** *Under equation (B.27), for every finite pair of query values (a, b) and every $M < \infty$, locally uniformly for $q \in [0, M]$,*

$$\Lambda_n(b_n q; a, b) \longrightarrow \lambda_{a,b}(q) := 4^p q^{2p} \frac{\varphi_p(b)}{\varphi_p(a)}, \tag{B.28}$$

and

$$\frac{M_{x,n}(b_n q; a, b)}{\sqrt{u_n}} \longrightarrow (2q)^p,\ \frac{M_{y,n}(b_n q; a, b)}{\sqrt{u_n}} \longrightarrow (2q)^p. \tag{B.29}$$

Proof. For the joint mean measure, start from the conditional competitor variables $(X_2, Z_2) = (x, z)$. Use the exact change of variables

$$s = \alpha_n(x - a),\ t = (1 - \delta_n)s + h_n(z - b).$$

Its Jacobians are $\mathrm{d}x = \alpha_n^{-p}\,\mathrm{d}s$ and $\mathrm{d}z = h_n^{-p}\,\mathrm{d}t$. Since $m_n/\alpha_n^p = 1/\varphi_p(a)$,

$$\Lambda_n(r; a, b) = \frac{1}{\varphi_p(a) h_n^p} \int_{C_r \times C_r} \varphi_p(a + s/\alpha_n)\varphi_p\left(b + \frac{t - (1 - \delta_n)s}{h_n}\right) \mathrm{d}s\ \mathrm{d}t. \tag{B.30}$$

Set $s = b_n\sigma$ and $t = b_n\tau$. The ratios appearing in the integrand satisfy

$$\left(\frac{b_n}{\alpha_n}\right)^p = \frac{\sqrt{u_n}}{m_n \varphi_p(a)} \longrightarrow 0, \tag{B.31}$$

$$\left(\frac{b_n}{h_n}\right)^p = \frac{\sqrt{u_n}}{m_n \varphi_p(a) \delta_n^{p/2} (1 - \delta_n)^{p/2}} \longrightarrow 0, \tag{B.32}$$

because

$$m_n \delta_n^{p/2} = \frac{m_n}{n} u_n$$

and $u_n \to \infty$. Moreover,

$$\frac{u_n}{h_n^p} = \frac{n}{m_n \varphi_p(a)(1 - \delta_n)^{p/2}} \longrightarrow \frac{1}{\varphi_p(a)}. \tag{B.33}$$

Since $b_n^{2p} = u_n$, equation (B.30) becomes

$$\Lambda_n(b_n q; a, b) = \frac{u_n}{\varphi_p(a) h_n^p} \int_{C_q \times C_q} \varphi_p(a + (b_n/\alpha_n)\sigma) \times \varphi_p\left(b + \frac{b_n}{h_n}\{\tau - (1 - \delta_n)\sigma\}\right) \mathrm{d}\sigma\ \mathrm{d}\tau. \tag{B.34}$$

On every fixed box, the two densities in the integrand converge uniformly to $\varphi_p(a)$ and $\varphi_p(b)$, respectively. They are bounded by the finite global supremum of the Gaussian density, and the prefactor converges by equation (B.33). Dominated convergence gives

$$\Lambda_n(b_n q; a, b) \longrightarrow |C_q|^2 \frac{\varphi_p(b)}{\varphi_p(a)} = 4^p q^{2p} \frac{\varphi_p(b)}{\varphi_p(a)}.$$

For the source marginal, the same change of variables used in equation (B.13), followed by $s = b_n\sigma$, gives

$$\frac{M_{x,n}(b_n q; a, b)}{\sqrt{u_n}} = \frac{1}{\varphi_p(a)} \int_{C_q} \varphi_p(a + (b_n/\alpha_n)\sigma)\, \mathrm{d}\sigma \longrightarrow (2q)^p. \tag{B.35}$$

For the output marginal, use the density g_{δ_n} and query output $y_{1,n}(a, b)$ defined in the proof of lemma B.4:

$$\frac{M_{y,n}(b_n q; a, b)}{\sqrt{u_n}} = \frac{1}{\varphi_p(a)} \int_{C_q} g_{\delta_n}(y_{1,n} + (b_n/\alpha_n)\tau)\, \mathrm{d}\tau \longrightarrow (2q)^p. \tag{B.36}$$

Here $b_n/\alpha_n \to 0$, $\delta_n \to 0$, and $y_{1,n}(a, b) \to a$.

The three limits have been proved pointwise in q. The left sides are nondecreasing functions of q, and the limits are continuous. Applying lemma B.2 to each proves local-uniform convergence. □

Lemma B.7 (Uniform law for the marginal empirical counts)**.** *Under equation (B.27), conditionally on every finite (a, b), for each $M < \infty$,*

$$\sup_{0 \le q \le M} \left| \frac{1}{\sqrt{u_n}} \sum_{j=2}^{n} \mathbf{1}\{\mathsf{A}_{j,n} < b_n q\} - (2q)^p \right| \xrightarrow{\mathbb{P}} 0, \tag{B.37}$$

$$\sup_{0 \le q \le M} \left| \frac{1}{\sqrt{u_n}} \sum_{j=2}^{n} \mathbf{1}\{\mathsf{B}_{j,n} < b_n q\} - (2q)^p \right| \xrightarrow{\mathbb{P}} 0. \tag{B.38}$$

Proof. We prove the source statement; the output proof is identical. Define

$$\mathcal{N}_{x,n}(q) = \sum_{j=2}^{n} \mathbf{1}\{\mathsf{A}_{j,n} < b_n q\},\ \ \nu_{x,n}(q) = \mathbb{E}[\mathcal{N}_{x,n}(q) \mid a, b].$$

Continuity of the conditional distribution implies $\nu_{x,n}(q) = M_{x,n}(b_n q; a, b)$. For fixed M, lemma B.6 supplies a constant $K_M(a, b) < \infty$ such that, for all sufficiently large n,

$$\sup_{0 \le q \le M} \nu_{x,n}(q) \le K_M(a, b)\sqrt{u_n}.$$

At a fixed q, $\mathcal{N}_{x,n}(q)$ is binomial conditionally on (a, b), so

$$\mathrm{Var}(\mathcal{N}_{x,n}(q) \mid a, b) \le \nu_{x,n}(q) \le K_M(a, b)\sqrt{u_n}.$$

Fix $\epsilon > 0$. Since $f(q) = (2q)^p$ is uniformly continuous on $[0, M]$, choose a finite grid

$$0 = q_0 < q_1 < \cdots < q_L = M$$

such that

$$\max_{1 \le \ell \le L} \{f(q_\ell) - f(q_{\ell-1})\} < \epsilon.$$

Chebyshev's inequality and a union bound over this fixed grid give

$$\mathbb{P}\left(\max_{0\le\ell\le L}\frac{|\mathcal{N}_{x,n}(q_\ell)-\nu_{x,n}(q_\ell)|}{\sqrt{u_n}}>\epsilon\;\middle|\;a,b\right)\le\frac{(L+1)K_M(a,b)}{\epsilon^2\sqrt{u_n}}\longrightarrow 0. \tag{B.39}$$

By the local-uniform mean convergence in equation (B.29),

$$\sup_{0\le q\le M}\left|\frac{\nu_{x,n}(q)}{\sqrt{u_n}}-f(q)\right|\longrightarrow 0.$$

For $q\in[q_{\ell-1},q_\ell]$, monotonicity yields

$$\mathcal{N}_{x,n}(q_{\ell-1})\le\mathcal{N}_{x,n}(q)\le\mathcal{N}_{x,n}(q_\ell).$$

On the complement of the event in equation (B.39), the upper bound, divided by $\sqrt{u_n}$, is at most

$$f(q_\ell)+\epsilon+o(1)\le f(q)+2\epsilon+o(1),$$

and the corresponding lower bound is at least

$$f(q_{\ell-1})-\epsilon-o(1)\ge f(q)-2\epsilon-o(1).$$

Thus the supremum in equation (B.37) is at most $2\epsilon+o(1)$ with conditional probability tending to one. Let $\epsilon\downarrow 0$. This proves the source statement. Replacing A by B and using the second marginal mean limit proves equation (B.38). A union bound makes the two statements joint. □

Theorem B.8 (Regular $\sqrt{u_n}$-count law)**.** *Under equation (B.27), let*

$$U,V\overset{\text{iid}}{\sim}N(0,I_p),\;\;G_k\sim\text{Gamma}(k,1),$$

independently, and define

$$Q_{p,k}=\left\{\frac{G_k\varphi_p(U)}{4^p\varphi_p(V)}\right\}^{1/(2p)},\;\;L_{p,k}=(2Q_{p,k})^p=\left\{G_k\frac{\varphi_p(U)}{\varphi_p(V)}\right\}^{1/2}. \tag{B.40}$$

Then

$$\left(U,V,\frac{\rho_n}{b_n},\frac{N_{x,1}}{\sqrt{u_n}},\frac{N_{y,1}}{\sqrt{u_n}}\right)\Rightarrow(U,V,Q_{p,k},L_{p,k},L_{p,k}). \tag{B.41}$$

In particular, equation (24) holds,

$$\mathbb{P}(S_{1,n}=1)\longrightarrow 0,$$

and

$$\overline{S}_{n,k}\longrightarrow 0\;\text{ in }L^1.$$

Proof. Fix finite query values (a,b) and put $Q_n=\rho_n/b_n$. For each $q>0$, lemma B.6 gives

$$m_n\mathbb{P}(\mathsf{J}_{2,n}\le b_nq\mid a,b)\longrightarrow\lambda_{a,b}(q)=4^pq^{2p}\frac{\varphi_p(b)}{\varphi_p(a)}.$$

The binomial order-statistic identity and the Poisson limit therefore give

$$\begin{aligned}\mathbb{P}(Q_n \le q \mid a,b) &\longrightarrow \mathbb{P}\{\mathrm{Pois}(\lambda_{a,b}(q)) \ge k\} \\ &= \mathbb{P}\{G_k \le \lambda_{a,b}(q)\}. \end{aligned} \tag{B.42}$$

Equivalently,

$$Q_n \Rightarrow Q(a,b) := \left\{ \frac{G_k \varphi_p(a)}{4^p \varphi_p(b)} \right\}^{1/(2p)} \text{ conditionally on } (a,b). \tag{B.43}$$

In particular, Q_n is conditionally tight.

At the random radius Q_n, the identities equation (B.4) read

$$N_{x,1} = \mathcal{N}_{x,n}(Q_n),\ N_{y,1} = \mathcal{N}_{y,n}(Q_n),$$

where the empirical functions are those of lemma B.7. Fix $M < \infty$. On $\{Q_n \le M\}$, the uniform empirical law gives

$$\frac{N_{x,1}}{\sqrt{u_n}} - (2Q_n)^p \xrightarrow{\mathbb{P}} 0,\ \frac{N_{y,1}}{\sqrt{u_n}} - (2Q_n)^p \xrightarrow{\mathbb{P}} 0 \tag{B.44}$$

conditionally on (a,b). Conditional tightness of Q_n lets $M \uparrow \infty$, so equation (B.44) holds without the restriction $Q_n \le M$. Combining it with equation (B.43) and the continuous mapping theorem proves

$$\left(Q_n, \frac{N_{x,1}}{\sqrt{u_n}}, \frac{N_{y,1}}{\sqrt{u_n}} \right) \Rightarrow (Q(a,b), (2Q(a,b))^p, (2Q(a,b))^p)$$

conditionally on every finite (a,b).

The conditional law of G_k in this display does not depend on the query values, so G_k may be realized independently of (U,V). For any bounded continuous test function of the query values, radius, and two counts, the conditional expectation converges pointwise for every finite (a,b) and is uniformly bounded. Dominated convergence over $(U,V) \sim N(0,I_p)^{\otimes 2}$ proves the joint convergence equation (B.41).

The variable $L_{p,k}$ is strictly positive and finite almost surely. For $\epsilon > 0$, once $k/\sqrt{u_n} < \epsilon$, the portmanteau theorem and equation (24) imply

$$\limsup_n \mathbb{P}(N_{x,1} \le k) \le \limsup_n \mathbb{P}\left(\frac{N_{x,1}}{\sqrt{u_n}} \le \epsilon \right) \le \mathbb{P}(L_{p,k} \le \epsilon).$$

Letting $\epsilon \downarrow 0$ gives $\mathbb{P}(N_{x,1} \le k) \to 0$. Saturation forces $N_{x,1} \in \{k-1, k\}$, hence

$$\mathbb{P}(S_{1,n} = 1) \le \mathbb{P}(N_{x,1} \le k) \longrightarrow 0.$$

Finally, exchangeability and $0 \le \overline{S}_{n,k} \le 1$ give

$$\mathbb{E}|\overline{S}_{n,k}| = \mathbb{E}\overline{S}_{n,k} = \mathbb{P}(S_{1,n} = 1) \longrightarrow 0.$$

□

Remark B.9 (Why $\delta_n \to 0$ cannot be omitted). Suppose instead that $\delta_n \to \delta_0 \in (0,1)$. Repeating the exact calculation in equation (B.30) gives

$$\Lambda_n(b_n q; a, b) \longrightarrow 4^p q^{2p} \frac{\varphi_p(b)}{\varphi_p(a)(1-\delta_0)^{p/2}}.$$

The source marginal still has normalized mean $(2q)^p$, whereas the output marginal has normalized mean

$$(2q)^p \frac{g_{\delta_0}((1-\delta_0)a + \sqrt{\delta_0(1-\delta_0)}\, b)}{\varphi_p(a)}.$$

These two constants need not agree. Bare $u_n \to \infty$ therefore does not imply the common-limit law in equation (24); the shrinking-channel assumption $\delta_n \to 0$ changes the theorem rather than merely simplifying its proof.

C Score moments and replacement variance

We prove the exact-convention replacement variance bound and score-moment convergence used in theorem 4.1. The variance bound is uniform in the positive noise level and does not assume independent query scores.

C.1 A geometric reverse-neighbor bound

We first isolate the deterministic geometry used by the variance proof.

Lemma C.1 (Reverse nearest-neighbor bound)**.** *For every fixed d, there is a finite constant κ_d with the following property. Let $\mathcal{Z} \subset \mathbb{R}^d$ be finite and suppose the relevant distances are distinct. For a fixed $z \in \mathcal{Z}$, the number of centers $x \in \mathcal{Z} \setminus \{z\}$ for which z is among the first m max-norm nearest neighbors of x is at most $\kappa_d m$.*

Proof. Partition the max-norm unit sphere into finitely many sets of max-norm diameter strictly less than one, and extend each set radially to a cone. Such a finite partition exists by compactness. Suppose $a = r\theta$ and $b = s\eta$, where $0 < s \le r$, are in one cone and $\|\theta\|_\infty = \|\eta\|_\infty = 1$. Then

$$\|a - b\|_\infty < r. \tag{C.1}$$

Indeed, put $t = s/r$. If $|\theta_j| < 1$, convexity of $t \mapsto |\theta_j - t\eta_j|$ bounds it by

$$\max\{|\theta_j|, |\theta_j - \eta_j|\} < 1.$$

If $|\theta_j| = 1$, diameter less than one forces η_j to have the same sign and to be nonzero, and hence $|\theta_j - t\eta_j| < 1$ for every $0 < t \le 1$. This proves equation (C.1).

Fix a cone and order its candidate centers by increasing distance from z. Every earlier center in that cone is closer to a later center than z is, by equation (C.1). Thus z cannot be among the first m neighbors of the $(m+1)$st or any later center in that ordering. There are at most m qualifying centers per cone. Taking κ_d to be the finite number of cones proves the claim. □

Lemma C.2 (Harmonic reverse-rank bound)**.** *Suppose a fixed point z is among the first m_i nearest neighbors of each center indexed by $i \in A$, where $1 \le m_i \le n-1$. Then*

$$\sum_{i \in A} \frac{1}{m_i} \le 2\kappa_d\{1 + \lceil \log_2 n \rceil\}. \tag{C.2}$$

Proof. Group the indices according to $2^\ell \le m_i < 2^{\ell+1}$. By lemma C.1, a group has at most $\kappa_d 2^{\ell+1}$ elements, while every summand in it is at most $2^{-\ell}$. There are at most $1 + \lceil \log_2 n \rceil$ nonempty groups, proving equation (C.2). □

C.2 A Variance Bound Independent of the Sampling Law

Theorem C.3 (Exact strict-radius KSG variance)**.** *Fix p, k, and let $n > k$. For an iid sample from any law having a Lebesgue density on $\mathbb{R}^p \times \mathbb{R}^p$, evaluated with the query-excluded, strict-radius KSG-1 convention in equations (3) and (4),*

$$\operatorname{Var}(\widehat{I}_{n,k}) \le C_{p,k} \frac{(1+\log n)^2}{n}, \tag{C.3}$$

where $C_{p,k} < \infty$ is independent of the sampling law. The same constant therefore applies uniformly to all sample sizes and laws in the shrinking-noise Gaussian triangular array.

Proof. For a data set $\mathcal{W} = (W_1, \dots, W_n)$, write

$$Q(\mathcal{W}) = \sum_{i=1}^{n} \{\psi(N_{x,i}+1) + \psi(N_{y,i}+1)\}.$$

Replace W_j by an independent copy W_j', producing $\mathcal{W}^{(j)}$. We first bound $|Q(\mathcal{W}) - Q(\mathcal{W}^{(j)})|$ on the probability-one event on which, in both samples, the distances from each query are distinct in the joint space and in each marginal space, and every joint boundary point is active in exactly one block. These exclusions follow from the joint Lebesgue density: after selecting coordinates and signs attaining the max norms, each excluded equality lies in a nontrivial affine hyperplane. A finite union over queries, competitors, and the two samples is still null. Marginal ties are excluded explicitly because marginal ranks are used below.

For a query $i \neq j$ common to both samples, its joint kth radius can change only if the old point W_j is among its first k joint neighbors in $\mathcal{W}$, or the new point W_j' is among its first k joint neighbors in $\mathcal{W}^{(j)}$. Applying lemma C.1 in dimension $2p$, there are at most $2\kappa_{2p}k$ such queries.

General position and the strict, query-excluded convention give

$$k - 1 \le N_{x,i}, N_{y,i} \le n - 1.$$

Thus each neighborhood's two-digamma summand lies in $[2\psi(k), 2\psi(n)]$. The replaced observation itself and all queries whose joint radius changes contribute at most

$$2(1 + 2\kappa_{2p}k)\{\psi(n) - \psi(k)\} \tag{C.4}$$

to the absolute change in Q.

Consider now the other queries, whose joint radii are unchanged. In a given marginal block, if the old and new points are both inside or both outside the strict ball, the count does not change. If only the old point is inside, write m for the original count. Its removal changes $\psi(m+1)$ to $\psi(m)$, an exact change of $1/m$. The old point has marginal rank at most m in the original sample, so lemma C.2 bounds the sum of these changes by

$$2\kappa_p\{1 + \lceil \log_2 n \rceil\}.$$

If only the new point is inside, write m for the final count. The increment is again $1/m$, and the new point has rank at most m in the replacement sample. The identical reverse-rank bound therefore applies to insertion, and separately to the x- and y-blocks. The total contribution of all queries with unchanged radii is therefore at most four times this display. Combining it with equation (C.4) and using $\psi(n) - \psi(k) \le C_k + \log n$, we obtain the deterministic bound

$$|Q(\mathcal{W}) - Q(\mathcal{W}^{(j)})| \le C'_{p,k}(1 + \log n). \tag{C.5}$$

The two deterministic digamma terms in equation (4) do not change. The independent-replacement Efron–Stein inequality [2], followed by equation (C.5), gives

$$\begin{aligned}\operatorname{Var}(\widehat{I}_{n,k}) &\le \frac{1}{2}\sum_{j=1}^{n}\mathbb{E}\left[\left\{\frac{Q(\mathcal{W})-Q(\mathcal{W}^{(j)})}{n}\right\}^2\right]\\ &\le \frac{(C'_{p,k})^2}{2}\frac{(1+\log n)^2}{n}.\end{aligned}$$

This proves equation (C.3). □

Remark C.4 (Why a plain bounded-degree argument is insufficient). Replacing one observation changes only $O_{p,k}(1)$ joint kth radii, but that observation can belong to many projected marginal balls. The reciprocal digamma increment and the harmonic reverse-rank bound control these additional score changes. This is also why the proof retains the strict-radius and $+1$ conventions explicitly.

C.3 Critical score moments

Let $(N_{x,n}, N_{y,n})$ denote the strict counts for a uniformly selected query and set

$$\ell_n = \psi(k) - \psi(N_{x,n}+1) - \psi(N_{y,n}+1).$$

With (M_x, M_y) as in proposition 3.1, define

$$\ell^{(u)} = \psi(k) - \psi(M_x+1) - \psi(M_y+1).$$

Lemma C.5 (Critical score moment convergence)**.** *If $u_n \to u \in (0,\infty)$, then*

$$\mathbb{E}\ell_n \longrightarrow \mathbb{E}\ell^{(u)},\ \mathbb{E}[\ell_n^2] \longrightarrow \mathbb{E}[(\ell^{(u)})^2]. \tag{C.6}$$

Proof. Theorem A.1 gives count convergence in the discrete count space. Applying its score map gives convergence in distribution of ℓ_n. By corollary A.12, the scores and their squares are uniformly integrable and have moments of every fixed finite order bounded uniformly in n. Truncation at a fixed score magnitude, followed by uniform integrability as the truncation increases, gives both limits in equation (C.6). □

C.4 Relation to prior KSG variance results

Zhao and Lai [13] state an $O((\log n)^2/n)$ variance rate for continuous distributions; their proof refers to the fixed-k analysis of Gao et al. [4], and the untruncated extension is not written out there in the exact strict-radius convention used here. We therefore do not claim the rate in theorem C.3 as novel. Its role is to make the convention and the required uniformity explicit: $C_{p,k}$ contains no density bound, derivative bound, support parameter, or factor depending on δ_n. In contrast, the fixed-law bias and consistency results of Gao et al. [4], Zhao and Lai [13] cannot simply be applied to the present triangular array, because their regularity constants need not remain uniform as $\delta_n \downarrow 0$.

D Overlap and integrability estimates for the bias endpoints

We prove the overlap, logarithmic integrability, and Poisson-score estimates used in theorem 5.1, all for the limiting Poisson functional.

D.1 A common overlap representation

Let $A, B \overset{\text{iid}}{\sim} N(0, I_p)$, let $G \sim \text{Gamma}(k, 1)$ be independent, and put

$$h = \{u\varphi_p(A)\}^{1/p}.$$

Following proposition 3.1, let R be the unique solution of

$$\Lambda_{h,B}(R) = \prod_{j=1}^{p} \mathbb{E}(2R - h|W_j - B_j|)_+ = G,$$

and define

$$M = (2R)^p, \ \mu = M - G. \tag{D.1}$$

Conditional on (A, B, G), the two marginal-only Poisson counts are independent with common mean μ. It is strictly positive almost surely for every $u > 0$. Indeed, with $a = 2R/h$,

$$\frac{G}{M} = \prod_{j=1}^{p} \frac{\ell_{B_j}(a)}{a},$$

and every factor is strictly less than one because $\ell_v(a)/a = \mathbb{E}(1 - |W - v|/a)_+$ and a nondegenerate Gaussian displacement is nonzero almost surely. Hence $G/M < 1$ and $\mu = M - G > 0$, so all logarithms below are well defined.

For later use, define, for $a > 0$ and $v \in \mathbb{R}$,

$$\ell_v(a) = \mathbb{E}(a - |W - v|)_+ = \int_{-a}^{a} (a - |z|)\varphi(v + z)\, \mathrm{d}z. \tag{D.2}$$

With $a = 2R/h$, the radius equation and marginal volume are

$$G = h^p \prod_{j=1}^{p} \ell_{B_j}(a), \ M = (ha)^p. \tag{D.3}$$

Recall from equations (16) and (17) that

$$g_m(\mu) = \mathbb{E}\psi\{m + \text{Pois}(\mu)\}, \ B_{p,k}(u) = \log u + \psi(k) - \mathbb{E}\{g_k(\mu) + g_{k+1}(\mu)\}.$$

D.2 The thin endpoint, including the $O(u^{1/p})$ remainder

Lemma D.1 (Expected marginal-only mass at small u)**.** *For $0 < u \le 1$,*

$$\mathbb{E}\mu \le C_{p,k} u^{1/p}. \tag{D.4}$$

Proof. For a deterministic displacement $z \in \mathbb{R}^p$, a telescoping product bound gives

$$(2R)^p - \prod_{j=1}^{p} (2R - h|z_j|)_+ \le h(2R)^{p-1} \|z\|_1. \tag{D.5}$$

To verify it, replace the p factors one at a time and bound every unchanged factor by $2R$. Each coordinate loss is at most $\min\{h|z_j|, 2R\}$. If a coordinate displacement exceeds $2R$, its single term on the right of equation (D.5) is already at least $(2R)^p$, so the same inequality still holds.

Average equation (D.5) over W, evaluate it at the random solution R, and set

$$L(B) = \mathbb{E}\|W - B\|_1 \le C_p + \|B\|_1, \ q = (p-1)/p.$$

Using $M = G + \mu$, we obtain

$$\mu = M - \Lambda_{h,B}(R) \le hL(B)M^q = hL(B)(G+\mu)^q. \tag{D.6}$$

For $p = 1$, this directly gives $\mu \le hL(B)$. For $p > 1$, concavity and Young's inequality yield

$$\begin{aligned}\mu &\le hL(B)G^q + hL(B)\mu^q \\ &\le hL(B)G^q + \frac{1}{2}\mu + C_p\{hL(B)\}^p.\end{aligned}$$

Thus, in every dimension,

$$\mu \le C_p\left[hL(B)G^{(p-1)/p} + \{hL(B)\}^p\right]. \tag{D.7}$$

Since $h = u^{1/p}\varphi_p(A)^{1/p}$, the Gaussian and Gamma moments on the right are finite. For $0 < u \le 1$, the first term has expectation $O(u^{1/p})$, and the second has expectation $O(u)$, which is no larger. This proves equation (D.4). □

D.3 The regular endpoint: almost-sure overlap scaling

Lemma D.2 (Large-u scaling of the marginal-only mean)**.** *Under the coupling above,*

$$\frac{M}{\sqrt{u}} \longrightarrow Q := \left\{\frac{G\varphi_p(A)}{\varphi_p(B)}\right\}^{1/2}, \ \frac{\mu}{\sqrt{u}} \longrightarrow Q, \tag{D.8}$$

almost surely as $u \to \infty$.

Proof. For each fixed v, the triangular kernel in equation (D.2) has total mass a^2, and continuity of the Gaussian density gives

$$\frac{\ell_v(a)}{a^2} \longrightarrow \varphi(v), \ a \downarrow 0. \tag{D.9}$$

The product on the right of

$$\frac{G}{h^p} = \prod_{j=1}^{p} \ell_{B_j}(a)$$

is continuous and strictly increasing from zero to infinity. Since $G/h^p = G/\{u\varphi_p(A)\} \to 0$ almost surely, its solution $a = a_u$ tends to zero. Equations (D.3) and (D.9) then give

$$G = \frac{M^2}{h^p}\varphi_p(B)\{1 + o(1)\},$$

which proves the first convergence in equation (D.8). The second follows from $\mu = M - G$ and $G/\sqrt{u} \to 0$. □

D.4 Uniform integrability of the logarithmic scaling

Pointwise convergence in lemma D.2 is not enough for the bias, because A, B, G are unbounded and the score is logarithmic. The next lemma supplies the needed envelope.

Lemma D.3 (Logarithmic moment convergence)**.** *As $u \to \infty$,*

$$\mathbb{E} \log \frac{\mu}{\sqrt{u}} \longrightarrow \frac{1}{2}\mathbb{E}\{\log G + \log \varphi_p(A) - \log \varphi_p(B)\}. \tag{D.10}$$

Equivalently,

$$2\mathbb{E} \log \mu - \log u \longrightarrow \psi(k). \tag{D.11}$$

Proof. We first control $\log(M/\sqrt{u})$, then show that replacing M by $\mu = M - G$ does not change the limiting expectation. Let $\varphi_0 = \varphi(0)$. From equation (D.2),

$$\ell_v(a) \le a^2 \varphi_0. \tag{D.12}$$

For a lower bound, set $b_0 = \min(a/2, 1/2)$. On $|z| \le b_0$, we have $a - |z| \ge a/2$ and

$$\varphi(v + z) \ge \varphi(v) e^{-|v|/2 - 1/8}.$$

Therefore, with

$$c(v) = \frac{1}{2}\varphi(v) e^{-|v|/2-1/8}, \; C(B) = \prod_{j=1}^{p} c(B_j),$$

$$\ell_v(a) \ge \begin{cases} c(v)a^2, & 0 < a \le 1, \\ c(v)a, & a > 1. \end{cases} \tag{D.13}$$

Combine equations (D.3) and (D.12) to obtain

$$\frac{M}{\sqrt{u}} \ge \left\{ \frac{G\varphi_p(A)}{\varphi_0^p} \right\}^{1/2}.$$

If $a \le 1$, equation (D.13) gives

$$\frac{M}{\sqrt{u}} \le \left\{ \frac{G\varphi_p(A)}{C(B)} \right\}^{1/2};$$

if $a > 1$, it gives

$$\frac{M}{\sqrt{u}} \le \frac{G}{C(B)\sqrt{u}} \le \frac{G}{C(B)} \; (u \ge 1).$$

Hence, for all $u \ge 1$,

$$\left\{ \frac{G\varphi_p(A)}{\varphi_0^p} \right\}^{1/2} \le \frac{M}{\sqrt{u}} \le \left\{ \frac{G\varphi_p(A)}{C(B)} \right\}^{1/2} + \frac{G}{C(B)}. \tag{D.14}$$

The logarithms of the two envelopes are bounded in absolute value by a constant times

$$1 + \|A\|_2^2 + \|B\|_2^2 + \|B\|_1 + |\log G| + \log(1 + G). \tag{D.15}$$

This variable has moments of every finite order. Thus $\{\log(M/\sqrt{u}) : u \ge 1\}$ is uniformly integrable, and lemma D.2 implies

$$\mathbb{E} \log \frac{M}{\sqrt{u}} \longrightarrow \mathbb{E} \log Q. \tag{D.16}$$

It remains to replace M by $\mu = M - G$. Define

$$\vartheta = \frac{\mu}{M} = 1 - \prod_{j=1}^{p} \frac{\ell_{B_j}(a)}{a}.$$

For one coordinate,

$$1 - \frac{\ell_v(a)}{a} = \mathbb{E} \min\{|W - v|/a, 1\}. \tag{D.17}$$

Since $1 - \prod_j q_j \geq 1 - q_1$ when $0 \leq q_j \leq 1$, Gaussian unimodality and equation (D.17) give

$$\vartheta \geq c_0 := \mathbb{P}(|W| \geq 1) > 0 \ (a \leq 1).$$

For completeness, the intermediate bound is

$$\vartheta \geq \mathbb{P}(|W - B_1| \geq a) \geq \mathbb{P}(|W| \geq a) \geq \mathbb{P}(|W| \geq 1).$$

The middle inequality holds because, among all intervals of the fixed length $2a$, the interval centered at the mode of the standard Gaussian has the largest probability mass; the last uses $a \leq 1$. For $a > 1$,

$$\vartheta \geq \frac{d(B_1)}{a}, \ d(v) = \mathbb{E}[|W - v|\mathbf{1}\{|W - v| \leq 1\}] \geq \frac{1}{4}\varphi(|v| + 1). \tag{D.18}$$

On the same event $a > 1$, the second line of equation (D.13) and the radius equation give

$$a^p \leq \frac{G}{u\varphi_p(A)C(B)} \leq \frac{G}{\varphi_p(A)C(B)} \ (u \geq 1). \tag{D.19}$$

It follows from equations (D.18) and (D.19) that $(-\log \vartheta)_+$ is also bounded by an integrable random variable of the form equation (D.15), uniformly over $u \geq 1$. Because $a_u \to 0$, $\vartheta_u \to 1$ almost surely. Dominated convergence and equation (D.16) now yield

$$\mathbb{E} \log \frac{\mu}{\sqrt{u}} = \mathbb{E} \log \frac{M}{\sqrt{u}} + \mathbb{E} \log \vartheta \longrightarrow \mathbb{E} \log Q,$$

which is equation (D.10).

Finally, A and B are iid, so their expected log-density terms cancel. Differentiating the Gamma integral shows $\mathbb{E} \log G = \psi(k)$. Multiplying equation (D.10) by two proves equation (D.11). □

D.5 The Poisson–digamma correction

Lemma D.4 (Exact Poisson–digamma identity)**.** *If $N \sim \mathrm{Pois}(\mu)$, with $\mu > 0$, then*

$$\mathbb{E}\psi(N + 1) = \log \mu + E_1(\mu), \ E_1(\mu) = \int_{\mu}^{\infty} \frac{e^{-t}}{t} \, \mathrm{d}t. \tag{D.20}$$

Consequently, for every integer $m \geq 1$,

$$g_m(\mu) - \log \mu = E_1(\mu) + \sum_{j=1}^{m-1} \mathbb{E} \frac{1}{N + j}. \tag{D.21}$$

The right side tends to zero as $\mu \to \infty$ and is bounded by

$$C_m + (-\log \mu)_+. \tag{D.22}$$

Proof. Using $\psi(N+1) = H_N - \gamma$ and

$$H_N = \int_0^1 \frac{1-t^N}{1-t}\,\mathrm{d}t,$$

Tonelli's theorem and $\mathbb{E}t^N = e^{-\mu(1-t)}$ give

$$\mathbb{E}H_N = \int_0^\mu \frac{1-e^{-s}}{s}\,\mathrm{d}s = \gamma + \log\mu + E_1(\mu).$$

This proves equation (D.20). The recurrence

$$\psi(N+m) = \psi(N+1) + \sum_{j=1}^{m-1} \frac{1}{N+j}$$

proves equation (D.21).

For $\mu \le 1$,

$$E_1(\mu) \le -\log\mu + \int_1^\infty e^{-t}/t\,\mathrm{d}t,$$

and the reciprocal terms are bounded by $\sum_{j<m} 1/j$. For $\mu \ge 1$, all terms are bounded, and each tends to zero as $\mu \to \infty$. For the reciprocal terms, couple N_μ as the count of a unit-rate Poisson process on $[0,\mu]$. Then $N_\mu \uparrow \infty$ almost surely, and dominated convergence applies to $1/(N_\mu + j)$. These observations give both the limit and equation (D.22). □

E Saturation, Bias Correction, and the Gaussian Comparator

We establish the saturation variance and bias continuity used in corollaries 6.1 and 6.2, then give sufficient numerical-lookup conditions and an exact Gaussian parametric comparator.

E.1 Variance of the empirical fraction

Lemma E.1 (Distribution-uniform saturation variance)**.** *For an iid sample of size $n > k$ from any law having a Lebesgue density on $\mathbb{R}^p \times \mathbb{R}^p$, evaluated with the strict, query-excluded convention equation (3),*

$$\mathrm{Var}(\overline{S}_{n,k}) \le \frac{D_{p,k}^2}{2n},\ D_{p,k} = 1 + 2\kappa_{2p}k + 4\kappa_p(k+1), \tag{E.1}$$

where κ_d is the geometric constant in lemma C.1. In particular, the bound is independent of the sampling density.

Proof. Write $T(\mathcal{W}) = \sum_{i=1}^n S_{i,n}$, and replace observation W_j by an independent copy W_j', giving $\mathcal{W}^{(j)}$. Work on the probability-one general-position event used in theorem C.3, including absence of marginal distance ties in both samples.

As in the variance proof for KSG, a common query's joint kth radius can change only if the old point is among its first k joint neighbors in $\mathcal{W}$, or the new point is among its first k joint neighbors in $\mathcal{W}^{(j)}$. Lemma C.1 bounds the number of such queries by $2\kappa_{2p}k$. Together with the replaced query itself, they account for at most $1 + 2\kappa_{2p}k$ changed indicators.

Consider any remaining query. Its radius is unchanged and each marginal count changes by at most one. If its saturation indicator changes, one of the two configurations is saturated. Both

counts in that configuration are at most k, and both counts in the other are therefore at most $k+1$. At least one marginal count changes. The old or new point responsible for that change belongs to the corresponding strict marginal ball in one of the two samples, whose count is at most $k+1$. Its marginal nearest-neighbor rank in that sample is consequently at most $k+1$. Applying lemma C.1 to the old and new points in each of the two marginal spaces bounds the number of these radius-stable queries by $4\kappa_p(k+1)$. A union bound suffices even if a query is included more than once.

Each indicator changes by at most one, so

$$|T(\mathcal{W}) - T(\mathcal{W}^{(j)})| \le D_{p,k}.$$

The independent-replacement Efron–Stein inequality gives

$$\mathrm{Var}(\overline{S}_{n,k}) \le \frac{1}{2}\sum_{j=1}^{n}\mathbb{E}\left[\left\{\frac{T(\mathcal{W}) - T(\mathcal{W}^{(j)})}{n}\right\}^2\right] \le \frac{D_{p,k}^2}{2n}.$$

This proves equation (E.1). □

E.2 Continuity and integrability of the bias curve

Lemma E.2 (Continuity of the critical bias)**.** *For fixed p, k, $B_{p,k}$ is finite and continuous on $(0,\infty)$, and hence bounded and uniformly continuous on every compact subinterval.*

Proof. First, for integers $m \ge 1$ and $0 \le s \le t$, couple $P_s \sim \mathrm{Pois}(s)$ and $P_t = P_s + Q$, where $Q \sim \mathrm{Pois}(t-s)$ is independent. The digamma recurrence gives

$$0 \le \psi(m+P_t) - \psi(m+P_s) = \sum_{j=0}^{Q-1}\frac{1}{m+P_s+j} \le \frac{Q}{m}.$$

Every term is integrable, since $|\psi(m+P_s)| \le |\psi(m)| + P_s/m$. Taking expectations proves

$$|g_m(t) - g_m(s)| \le \frac{|t-s|}{m},\ |g_m(t)| \le |\psi(m)| + \frac{t}{m}. \tag{E.2}$$

In particular, g_m is continuous.

Next, fix $0 < a < b < \infty$ and use the common coupling (A, B, G_k) from the proof of corollary 6.1, which gives continuous paths $u \mapsto \mu_u$, increasing in u. The pointwise estimate equation (D.7) was obtained before restricting $u \le 1$; thus it holds at $u = b$ as well. Since $\varphi_p(A) \le (2\pi)^{-p/2}$, it implies

$$0 \le \mu_u \le \mu_b \le C_p\left[b^{1/p}(C_p + \|B\|_1)G_k^{(p-1)/p} + b(C_p + \|B\|_1)^p\right],\ u \in [a,b].$$

The right-hand side is integrable: all Gaussian moments are finite, the indicated nonnegative Gamma moment is finite, and the query values and arrival are independent. Consequently $\mathbb{E}\mu_b < \infty$.

If $u_j \to u$ within this interval, pathwise continuity and equation (E.2) give $g_m(\mu_{u_j}) \to g_m(\mu_u)$ almost surely, dominated by $|\psi(m)| + \mu_b/m$. Dominated convergence proves continuity of both expectations in equation (16). Together with continuity of $\log u$, this proves continuity and finiteness of $B_{p,k}$. Every point of $(0,\infty)$ lies in such a compact interval. Boundedness and uniform continuity on compact intervals then follow from continuity. □

E.3 What numerical lookup tables must satisfy

Proposition E.3 (A sufficient deterministic approximation condition)**.** *Under the assumptions of corollary 6.2, let $q_n : K \to \mathbb{R}$ be continuous and strictly decreasing, and let $B_n : K \to \mathbb{R}$ be measurable deterministic functions. Suppose*

$$\epsilon_n = \sup_{v \in K} |q_n(v) - q(v)| \longrightarrow 0, \ \eta_n = \sup_{v \in K} |B_n(v) - B(v)| \longrightarrow 0. \tag{E.3}$$

Set

$$\widetilde{u}_n = q_n^{-1}\big(\Pi_{[q_n(b),q_n(a)]}(\overline{S}_{n,k})\big)\,.$$

Then $\widetilde{u}_n \to u$ and $\widehat{I}_{n,k} - B_n(\widetilde{u}_n) - I_n \to 0$, both in L^2.

Proof. If two closed intervals have corresponding endpoints within ϵ, their clipping maps differ by at most ϵ at every real argument. This follows directly from $\Pi_{[c,d]}(s) = \min\{d, \max\{c, s\}\}$: clipping is nondecreasing in each endpoint, and shifting both endpoints by $\pm\epsilon$ changes its value by at most ϵ. Apply this fact to $J_n = [q_n(b), q_n(a)]$ and $J = [q(b), q(a)]$. Since projection onto an interval is 1-Lipschitz and $q(u) \in J$,

$$\begin{aligned} |q(\widetilde{u}_n) - q(u)| &\le |q(\widetilde{u}_n) - q_n(\widetilde{u}_n)| + |\Pi_{J_n}(\overline{S}_{n,k}) - \Pi_J(\overline{S}_{n,k})| \\ &\quad + |\Pi_J(\overline{S}_{n,k}) - \Pi_J(q(u))| \\ &\le 2\epsilon_n + |\overline{S}_{n,k} - q(u)|. \end{aligned}$$

The right-hand side tends to zero in L^2. Uniform continuity of $q^{-1} : J \to K$, followed by the bounded splitting argument in the proof of corollary 6.2, gives $\widetilde{u}_n \to u$ in L^2. The same argument gives $B(\widetilde{u}_n) \to B(u)$ in L^2. Moreover,

$$|B_n(\widetilde{u}_n) - B(u)| \le \eta_n + |B(\widetilde{u}_n) - B(u)|.$$

This tends to zero in L^2, and the final decomposition in the proof of corollary 6.2 completes the proof. □

This proposition is a sufficient condition, not a certification of a fixed Monte Carlo grid. For example, exact values of q and B interpolated linearly on grids whose maximum spacing tends to zero satisfy equation (E.3) by uniform continuity; strict decrease of q makes its piecewise-linear interpolant strictly decreasing. Uniformly vanishing errors at the nodes preserve convergence of the interpolants, but an approximate q_n must still be made strictly decreasing. A fixed grid leaves interpolation and integration error. Independent simulation seeds help detect Monte Carlo sensitivity, but do not establish the uniform premise in equation (E.3). Nothing here permits the window to expand with n, provides an endpoint inversion guarantee, or constructs finite-sample confidence intervals for I_n.

E.4 An exact Gaussian parametric comparator

When the channel family is known, its residual variance gives a direct information estimate. The following standard Gaussian calculation makes that advantage explicit; it is a benchmark, not a novelty claim.

Proposition E.4 (Residual-variance comparator)**.** *For the Gaussian channel equation (1), with iid observations, define*

$$\widehat{\delta}_n = \frac{1}{np}\sum_{i=1}^{n} \|X_i - Y_{i,n}\|_2^2,\ \nu = np,\ c_\nu = \psi(\nu/2) - \log(\nu/2).$$

Then $\nu\widehat{\delta}_n/\delta_n \sim \chi^2_\nu$*. The untruncated plug-in estimator* $\widehat{I}_n^{\mathrm{plug}} = -(p/2)\log\widehat{\delta}_n$ *has mean error* $-pc_\nu/2$*. Its corrected version*

$$\widehat{I}_n^{\mathrm{G}} = -\frac{p}{2}\log\widehat{\delta}_n + \frac{p}{2}c_\nu \tag{E.4}$$

is exactly unbiased for I_n*, with*

$$\mathbb{E}(\widehat{I}_n^{\mathrm{G}} - I_n)^2 = \frac{p^2}{4}\psi_1(\nu/2) \le \frac{p}{2n} + \frac{1}{n^2}, \tag{E.5}$$

where ψ_1 *is the trigamma function. For fixed* p*, this implies* L^2 *consistency along every sequence* $0 < \delta_n < 1$*, with no restriction on* u_n*.*

Proof. The difference for observation i is

$$X_i - Y_{i,n} = \delta_n X_i - \sqrt{\delta_n(1-\delta_n)}Z_i.$$

It is a centered Gaussian vector with covariance $\{\delta_n^2 + \delta_n(1-\delta_n)\}I_p = \delta_n I_p$. Its coordinates are independent, as are the different observations. Thus the sum of all np squared standardized residuals is χ^2_ν, proving the claimed distribution.

Write $z = \nu/2$. If $T \sim \mathrm{Gamma}(z,1)$, the chi-square identity yields $\widehat{\delta}_n/\delta_n \overset{d}{=} T/z$. For real $s > -z$, integration of the Gamma density gives

$$\mathbb{E}T^s = \frac{\Gamma(z+s)}{\Gamma(z)}.$$

Differentiate twice at zero. The differentiations are justified by domination on any sufficiently small closed interval around zero: near zero, the integrands are bounded by a constant times $t^{z-1-\varepsilon}(1 + |\log t|^2)$, with $0 < \varepsilon < z$; near infinity, the exponential Gamma tail dominates the powers and logarithms. Consequently

$$\mathbb{E}\log T = \psi(z),\ \ \mathrm{Var}(\log T) = \psi_1(z).$$

It follows that $\mathbb{E}\log(T/z) = c_\nu$. Subtracting $I_n = -(p/2)\log\delta_n$ from the plug-in estimator proves its stated mean error, and adding $pc_\nu/2$ removes that error exactly. The variance is $p^2\psi_1(z)/4$.

Finally, the trigamma series and an integral comparison give

$$\psi_1(z) = \sum_{j=0}^{\infty}\frac{1}{(z+j)^2} \le \frac{1}{z^2} + \int_0^\infty \frac{\mathrm{d}t}{(z+t)^2} = \frac{1}{z^2} + \frac{1}{z}.$$

Substituting $z = np/2$ proves equation (E.5). The bound is independent of δ_n, so it applies to any allowed sequence of channel parameters. □

The comparison uses the known channel family and the standardized source covariance. It does not estimate information for an arbitrary joint Gaussian covariance matrix, let alone an unknown non-Gaussian law. The exact formulas concern the untruncated estimates, which can be negative in some finite samples. Projection of $\widehat{I}_n^{\mathrm{G}}$ onto $[0,\infty)$ cannot increase its squared error because $I_n > 0$, but would change the exact unbiasedness and variance identities. Neither version is a one-sided confidence bound.

F Numerical evaluation and experimental methods

F.1 Computing the Poisson Predictions

To compute the predictions, we draw source and noise values A, B and a Gamma arrival G_k, then solve for the joint radius. In the notation of Appendix D,

$$\ell_v(a) = \int_0^a (a-z)\{\varphi_1(v+z) + \varphi_1(v-z)\}\,\mathrm{d}z, \tag{F.1}$$
$$\Lambda_{h,B}(r) = h^p \prod_{j=1}^{p} \ell_{B_j}(2r/h).$$

The integrand is nonnegative. The reported calculations use a fixed upper radius w. If $G_k < \Lambda_{h,B}(w)$, bisection on $[0, w]$ determines R; otherwise the draw is recorded as unresolved. Appendix F.3 gives the search limits and their checks. The resolved radius gives $\mu_u = (2R)^p - G_k > 0$.

For saturation, one can average $e^{-2\mu_u}$ over the independent Monte Carlo draws. For bias, one may draw the Poisson marginal-only counts or integrate them out. Lemma D.4 gives

$$g_m(\mu) = \log\mu + E_1(\mu) + \sum_{j=1}^{m-1} \int_0^1 t^{j-1} e^{\mu(t-1)}\,\mathrm{d}t,\ \mu > 0. \tag{F.2}$$

Indeed, for $N \sim \mathrm{Pois}(\mu)$, Tonelli's theorem and the Poisson generating function give

$$\mathbb{E}\frac{1}{N+j} = \int_0^1 t^{j-1}\mathbb{E}t^N\,\mathrm{d}t = \int_0^1 t^{j-1}e^{\mu(t-1)}\,\mathrm{d}t.$$

At zero, use $g_m(0) = \psi(m)$. Near zero, direct summation of Poisson-weighted digamma values avoids cancellation between $\log\mu$ and $E_1(\mu)$.

F.2 Reducing Monte Carlo Noise

For the large-u calculation, we reduce Monte Carlo noise by subtracting a random quantity whose expectation is exactly zero. Let the bias score from one draw, in nats, be

$$T_u = \log u + \psi(k) - \psi(M_x + 1) - \psi(M_y + 1),$$

where the counts follow proposition 3.1. Its expectation is $B_{p,k}(u)$. The control Z in equation (34) has exact mean zero because

$$\mathbb{E}\log G_k = \frac{1}{\Gamma(k)}\int_0^\infty g^{k-1}e^{-g}\log g\,\mathrm{d}g = \frac{\Gamma'(k)}{\Gamma(k)} = \psi(k)$$

and $\mathbb{E}\|A\|_2^2 = \mathbb{E}\|B\|_2^2 = p$. Therefore $\mathbb{E}(T_u - Z) = B_{p,k}(u)$. Both the raw and adjusted scores are divided by $\log 2$ for bit units. The coefficient of Z is fixed at one rather than estimated from the draws. The raw and adjusted averages therefore estimate the same bias. The choice of Z follows the large-u limit in Appendix D.

F.3 Simulation Settings and Numerical Checks

The finite-sample study uses $p \in \{1, 2, 4\}$, $k \in \{3, 5, 10\}$, and $n \in \{250, 500, 1000, 2000, 4000, 8000\}$, with 30 independent repetitions for each dimension and sample size. Table 6 specifies every fixed-u setting. The polynomial paths use

$$\delta_n = n^{-2a/p}, \ a \in \{0.5, 0.75, 1, 1.25, 1.5, 2, 2.5\},$$

so $u_n = n^{1-a}$. These paths use source and noise samples independent of those in the fixed-u study.

Table 6: Fixed-u Gaussian simulation grid. Each listed exponent e specifies $u = 10^e$, at every sample size and neighbor index.

p	Exponents $e = \log_{10} u$
1	$-6, -5, -4, -3, -2.5, -2, -1.5, -1.25, -1,$ $-0.75, -0.5, -0.25, 0, 0.5, 1, 2$
2	$-9, -8, -7.5, -7, -6, -5, -4.5, -4, -3.5, -3,$ $-2.75, -2.5, -2.25, -2, -1.75, -1.5, -1, 0, 1, 2$
4	$-12, -11, -10, -9, -8, -7.5, -7, -6.5, -6,$ $-5.75, -5.5, -5.25, -5, -4.75, -4.5, -4, -3, -2, -1, 0, 1, 2$

The finite simulations generate independent standard Gaussian pairs (X_i, Z_i), form Y_i by equation (1), and compute the strict-radius counts in equation (3). For a fixed transition parameter, $\delta_n = (u/n)^{2/p}$, with $u < n$. Within each repetition, the same source and noise samples are reused across channel settings and neighbor indices. All simulations use Python, NumPy, and SciPy with double-precision arithmetic. Finite-sample counts use max-norm nearest-neighbor searches. At a positive joint radius ε_i, marginal searches use the largest representable radius below ε_i, ensuring that boundary points are excluded. The query itself is also excluded. We add no jitter and do not rescale either marginal. We check for zero radii, tied joint distances at the selected neighbor, and equal marginal distances for the boundary point; none occurred in the main study. No trials are discarded based on these checks.

The original Poisson reference uses $p \in \{1, 2, 4\}$ and $k \in \{3, 5, 10\}$. Its $\log_{10} u$ ranges are $[-4, 1]$, $[-7, 1]$, and $[-10, 0]$, respectively, with spacing 0.5. Each setting uses 30,000 independent draws of (A, B, G_k), an upper search radius $w = 24$, quadrature order 40, and 52 bisection steps. The radius is measured in the rescaled coordinates where each marginal has unit intensity. We reuse the draws across u within each (p, k). Accuracy checks start with the first 15,000 draws, $w = 12$, and quadrature order 20, then double each of these three settings in turn. Integer-decade nodes form the coarser comparison grid. The overlap is evaluated by Gaussian distribution-function identities when $2r/h > 1$ and by Gauss–Legendre quadrature of the nonnegative integral in equation (F.1) otherwise. The bias reference samples the conditional Poisson counts; the saturation reference averages $e^{-2\mu_u}$.

The dimension-four extension uses $p = 4, k = 5$ and $u = 10^a$ for $a = 0, 0.5, \ldots, 8$, with draws independent of the original Poisson study. It uses 30,000 independent draws, $w = 128$, quadrature order 40, and 52 bisection steps, checked against 15,000 draws, $w = 64$, and order 20. Draws are reused across u. The subsequent endpoint calculation uses two independent sets of 30,000 draws, $w = 512$, quadrature order 40, 56 bisection steps, and exponents $4, 6, 8, 10, 12$. The first set reuses the extension's draws to assess their Monte Carlo error; the second is generated independently.

All radii were resolved in the primary calculations used for the reported Poisson curves, endpoint calculation, and correction tables. Some draws in the smaller $w = 12$ check were unresolved. Its

bias averages use only resolved draws; saturation averages are bounded by assigning each unresolved draw first zero and then one. These smaller-radius checks are not used as the plotted reference.

A separate implementation check fixes $p = 1, k = 3, A = 0, B = 1.4$ and $h = 1.7$, hence $u = 1.7/\varphi_1(0)$. It compares 5,000 direct process simulations with 5,000 independent Gamma/count simulations, using $w = 8$, quadrature order 24, and 52 bisection steps for the latter. The direct method generates the union of the two marginal strips without truncating the Gaussian noise; all its radii are resolved. We compare boundary orientations, saturation frequencies, and mean scores. One Gamma/count radius is unresolved: its contribution to saturation is bounded as above, and its score is omitted from the mean. This is an implementation check at a fixed query.

Finite-point error bars are 95% Student-t intervals for setting means across 30 independent datasets. Poisson bands use mean plus or minus 1.96 standard errors across independent Monte Carlo draws. Probability intervals are clipped to $[0, 1]$. The intervals apply to individual settings, not the whole grid simultaneously. They describe sampling uncertainty; quadrature, radius-search, and finite-sample approximation errors are assessed separately by the checks above and the comparisons in Section 7.

F.4 Uncertainty in the Convergence Comparisons

The convergence comparisons reuse the simulations at all six sample sizes, $p = 1, 2, 4$, $u = 0.01, 0.1, 1$, and $k = 5$. The three u values are the common full-decade settings in the reference grids, selected before inspecting their residuals. This is a reanalysis of existing simulations, not a preregistered experiment. Both the mean-residual and mean-square analyses use all 30 repetitions at these settings. Within each dimension and sample size, the datasets are paired across u; the finite and Poisson simulations are independent. The bias reference averages sampled-count scores, while the saturation reference averages $e^{-2\mu}$. Residuals retain the finite-n term $\psi(n) - \log n$; it is not subtracted.

Let s_f be the standard error across 30 independent finite datasets, and s_P the standard error across 30,000 independent Monte Carlo draws from the Poisson law. The displayed residual interval has half-width

$$t_{0.975,\nu_W}\sqrt{s_f^2 + s_P^2},\ \ \nu_W = \frac{(s_f^2 + s_P^2)^2}{s_f^4/29 + s_P^4/29999}.$$

This approximate Welch–Satterthwaite interval combines the two independent sources of sampling error. Whole datasets, not individual neighborhoods, are the finite-sample replicates. The same Poisson reference is reused across sample sizes, so its error is shared across those comparisons. Likewise, settings computed from the same source and noise samples are paired. We therefore do not assign confidence intervals to summaries across settings by treating them as independent. Section 7 reports the results and the mean-square decomposition.

F.5 Implementing and Testing Bias Correction

We apply corollary 6.2 using two independently computed tables of the Poisson bias and saturation curves. Both are independent of the finite datasets and the main transition study. The first supplies the correction; the second checks sensitivity to Monte Carlo noise. Each table uses 30,000 independent draws of (A, B, G_k) per dimension, reused across its grid. We evaluate the Poisson-averaged scores g_m rather than simulate additional counts. The $\log_{10} u$ ranges are $[-1, 1]$, $[-2.5, 1]$, and $[-4.5, 0.5]$ for $p = 1, 2, 4$, with spacing 0.25.

The finite tests use all midpoints between adjacent $\log_{10} u$ grid values, at $n = 1000, 4000, 8000$ with $k = 5$ and 16 independent dataset repetitions per dimension and sample size. These datasets are independent of the main study; within a repetition, the source and noise samples are reused across u. We interpolate both curves linearly in $\log u$. To correct an estimate, we move any saturation fraction outside the table's range to the nearest endpoint, interpolate $\log u$ from that fraction, and subtract the bias evaluated at the resulting value.

The tables use $w = 64$, quadrature order 20, and 48 bisection steps. Checks at the endpoints and center use $w = 128$, order 40, and 56 steps with the same draws. Interpolation checks also reuse these draws, so differences reflect interpolation rather than a new Monte Carlo sample. Separately, we repeat every correction with the independently computed table to measure sensitivity to Monte Carlo noise.

A refinement check halves the $\log_{10} u$ spacing to 0.125, using the same two sets of 30,000 draws and all the original finite datasets. The original test points now lie on the refined grid. To check interpolation between grid points, we also evaluate five new midpoints spread across each dimension's range, using the same selection rule as for the original grid. The main results use the original tables; Section 7 reports the refinement comparison. This check changes the numerical approximation, not the finite datasets.

Mean-error intervals use 16 independent whole-dataset repetitions per setting. The RMSE figure instead uses percentile bootstrap intervals from 3,000 resamples of those repetitions, using the same resampled dataset indices for every method. Both types of interval hold the primary table fixed; they do not include uncertainty in its computed curves, which is checked separately using the second table. Datasets with zero saturation or an out-of-range saturation fraction remain in every reported comparison.